



NETWORK IMPLEMENTATION FOR GOLDEN HOTEL (PVT) LTD

**A dissertation submitted for the
Diploma in Network Engineering**

Group 16

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DECLARATION

We hereby declare that this project report titled “**Network Design and Implementation for Golden Hotel**” is an original work carried out by us under the supervision of **Mr.M.R.M.Ilham**, in partial fulfillment of the requirements for the award of the degree of **Diploma in Network Engineering** at **National Institute of Business Management School of Computing**.

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This report has not been submitted, either in whole or in part, for the award of any degree, diploma, or any other academic qualification at this or any other institution. All sources of information and materials used in the preparation of this report have been properly acknowledged and referenced.

We further declare that this work is free from plagiarism and represents our own efforts and understanding of the subject.

Certified by:

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Signature:

Date:

Acknowledgment

We would like to express our sincere gratitude to the National Institute of Business Management (NIBM), School of Computing, for providing us with the opportunity and facilities to successfully complete our project titled “Network Solution for Golden Hotel.” This project has been a valuable learning experience, allowing us to apply our theoretical knowledge to a practical, real-world scenario in the field of networking.

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Abstract

This project focuses on the design and implementation of a secure, scalable, and efficient computer network for **Golden Hotel**, a medium-sized hotel with 30 rooms and multiple service areas. The existing network lacked segmentation, security, redundancy, and centralized management.

The proposed solution introduces VLAN-based segmentation, dynamic IP addressing using DHCP, redundancy using HSRP, secure device management using SSH, access control using ACLs, wireless network separation for guests and staff, and virtualization using VMware. The network is designed and tested in a laboratory environment using industry-standard tools.

This project demonstrates the practical application of networking theories learned during the course to solve real-world organizational networking problems.

Table of Contents

Acknowledgment	3
Abstract	4
CHAPTER 1 - INTRODUCTION	12
1.1 Background of the Study	12
1.2 Organization Overview – Golden Hotel	13
1.3 Existing Network Scenario	13
1.4 Problem Statement	14
1.5 Aim of the Project	14
1.6 Objectives of the Project	14
1.7 Scope of the Project	15
1.8 Significance of the Project	15
1.9 Organization of the Report	15
CHAPTER 2 – Network Design	17
2.1 Overview of Network Design	17
2.2 Network Topology Design	17
2.2.1 Physical Network Topology.....	17
2.3 IP Addressing Design	18
2.3.1 IP Addressing Scheme	18
2.3.2 VLSM Design Table	18
2.4 Rack Layout and Equipment Placement	18
2.4.1 Network Equipment Rack on the Basement	19
2.4.2 Network Equipment Rack on the Ground Floor	20
2.4.3 Network Equipment Rack on 2,3,4 Floor	21
2.5 Floor Plan Network Design	22
2.5.1 Ground Floor Design	22
2.5.2 1st,2nd and 3rd Floor Design.....	23
2.5.3 4 th Floor Design (Rooftop).....	24
3.5.4 Basement floor	25
2.6 Wireless Network Design	25
2.6.1 Access Point Placement	25
2.6.2 Wi-Fi Coverage and Heat Map	25

2.7 Technologies Used and Implemented Protocols	29
2.8 Summary of Network Design	29
2.9 Bandwidth calculation	30
2.10 Bandwidth Calculation	31
CHAPTER 3 – IMPLEMENTATION	32
3.1 Introduction	32
3.2 Implementation Environment.....	32
3.3 Router Configuration	33
3.3.1 Basic Router Configuration	33
3.3.2 Network Address Translation (NAT)	36
3.4 Layer 3 Switch Configuration.....	38
3.4.1 VLAN Configuration	38
3.4.2 Inter-VLAN Routing.....	40
3.4.3 EtherChannel (Port Channel) Implementation.....	41
3.4.4 HSRP Configuration (Primary & Backup)	42
3.4.5 Access Control List (ACL) Configuration.....	42
3.5 Layer 2 Switch Configuration.....	43
3.5.1 Access and Trunk Ports.....	43
3.6 DHCP Server Implementation Using VMware	43
3.7 Wireless Network Implementation	46
3.8 Syslog and Remote Management.....	47
3.9 Summary.....	48
.....	49
CHAPTER 4 – EVALUATION.....	50
4.1 Introduction.....	50
4.2 VLAN Configuration Verification	50
4.2.1 Purpose of the Test.....	50
4.2.2 Test Method.....	51
4.2.3 Evaluation Outcome.....	51
4.3 Routing and NAT Evaluation.....	51
4.3.1 Purpose of the Test.....	51
4.3.2 Test Method.....	52

4.3.4 Evaluation Outcome.....	52
4.4 Secure Shell (SSH) Access Evaluation.....	52
4.4.1 Purpose of the Test.....	52
4.4.2 Test Method.....	52
4.4.3 Test Results	52
4.4.4 Evaluation Outcome.....	53
4.5 Syslog Server Evaluation.....	53
4.5.1 Purpose of the Test.....	53
4.5.2 Evaluation Outcome.....	53
4.6 DHCP Pool Utilization Evaluation	53
4.6.1 Purpose of the Test.....	53
4.6.2 Test Method.....	53
4.6.3 Evaluation Outcome.....	54
4.7 Redundancy and Failover Evaluation (HSRP)	54
4.7.1 Purpose of the Test.....	54
4.7.2 Test Method.....	54
4.7.3 Evaluation Outcome.....	54
4.8 Summary of Evaluation Results	55
.....	55
.....	58
CHAPTER 5 – CONCLUSION AND FUTURE WORK.....	59
5.1 Conclusion	59
5.2 Achievement of Project Objectives	60
5.3 Future Work	61
REFERENCES.....	62
ANNEXURE 1 – DEVICE SPECIFICATIONS.....	63
A1.1 Introduction	63
A1.1.1 Layer 2 Switch - Cisco Catalyst C9300 (C9300-48P-E)	63
A1.1.2 Firewall- Cisco Firepower 1120 (FPR1120-NGFW-K9).....	65
A1.1.3 Access Point - Cisco Catalyst 9120AXI-E (C9120AXI-E)	66
A1.1.4 WLAN Controller- Cisco Catalyst 9800 (C9800-L-C-K9)	68
A1.1.5 CCTV - Hikvision DS-9632NI-I8.....	70

A1.1.6 Rack Server - Dell PowerEdge (R430 1U E5-2603)	72
A1.1.7 Layer 3 Switch – Cisco Catalyst 9500	74
A1.1.8 Layer 1 switch – Cisco Catalyst 9200.....	76
A1.1.9 Types of Network Cables	78
A1.1.10 Desktop PC – Lenovo Yoga AIO 7	80
A1.1.11 Laptops – Lenovo ThinkPad X1 Carbon Gen11	81
A1.1.12 IP Phone – Cisco SPA 303	82
A2.2 Layer 3 Switch Specifications	83
A2.3 Layer 2 Switch Specifications	83
A2.4 Wireless Access Point Specifications	84
A2.5 Server Hardware Specifications	84
A2.6 Rack and Power Equipment	85
A2.7 Summary	85
A2.8 Summary of Project Outcomes	87
A2.9 Lessons Learned	88

List of Figures

Figure 1:Golden Hotel	12
Figure 2: Physical Network Diagram for Golden Hotel	17
Figure 3: Network Rack Diagram of Basement.....	19
Figure 4: Network Rack Diagram of Ground floor	20
Figure 5: Network Rack Diagram of 2,3,4 floor.....	21
Figure 6: Ground Floor design plan of the Golden Hotel	22
Figure 7: 1st,2nd & 3rd floor design plan of the Golden Hotel	23
Figure 8: 4th floor (Rooftop) design plan of the Golden Hotel	24
Figure 9: Basement design plan of the Golden Hotel	25
Figure 10: Wi-Fi heat map of ground floor golden hotel.....	26
Figure 11: Wi-Fi heat map of ground floor golden hotel.....	27
Figure 12: Wi-Fi heat map of 4th floor (rooftop) golden hotel.....	28
Figure 13: Wi-Fi heat map of Basement golden hotel.	29
Figure 14: host name configuration	33
Figure 15: IP addressing on router	34
Figure 16: configuring the SSH	35
Figure 17: configuring the SSH 2	35
Figure 18: SSH Configuration	35
Figure 19: SSH login to R1	36
Figure 20: NAT configuration.....	36
Figure 21: confirmation of NAT configuration.....	37
Figure 22: VLAN configuration in L3 switch	38
Figure 23: VLAN configuration in L3 switch	39
Figure 24: VLAN configuration in L3 switch	40
Figure 25: EtherChannel Configuration.....	41
Figure 26: HSRP enabled in core layer and the distribution layer.....	42
Figure 27: ACL	42
Figure 28: Trunk port & EtherChannel configuration	43
Figure 29: DHCP pools configured on the virtual server machine.....	43
Figure 30: DHCP pools configured on the virtual server machine.....	44
Figure 31: DHCP pools configured on the virtual server machine.....	44
Figure 32: DHCP Ip Addresses are assigned to the client machines accordingly.....	45
Figure 33: Windows Validation Server	46
Figure 34: SSH login	47
Figure 35: Syslog server Logs	47
Figure 36: physical Implementation at the network lab.....	48
Figure 37: physical Implementation at the network lab.....	49
Figure 38: vlan.....	51
Figure 39: Successful SSH login session.....	52
Figure 40: successful ping of the router port during implemantation.....	55

Figure 41: IP Addresses are obtained automatically with DHCP	56
Figure 42: Successful ping to Google server	57
Figure 43: DHCP configured Successfully	58
Figure 44: Cisco Catalyst C9300	63
Figure 45: Cisco Firepower 1120.....	65
Figure 46: Cisco Catalyst 9120AXI-E	66
Figure 47: Cisco Catalyst 9800.....	68
Figure 48: Hikvision DS-9632NI-I8	70
Figure 49: - Dell PowerEdge	72
Figure 50: Cisco Catalyst 9500.....	74
Figure 51: Cisco Catalyst 9200.....	76
Figure 52: Console cable	79
Figure 53: Desktop PC.....	80
Figure 54: Laptops	81
Figure 55: Ip Phones	82

List of Tables

Table 1: Ip Address Table.....	18
Table 2: Layer 3 Switch Specifications	83
Table 3: Layer 2 Switch Specifications	83
Table 4: Wireless Access Point Specifications.....	84
Table 5: Server Hardware Specifications.....	84
Table 6: Rack and Power Equipment.....	85
Table 7: Estimated Budget	86

List of Abbreviations

- IPv4 - Internet Protocol Version 4
- MAC - Media Access Control
- LAN - Local Area Network
- WAN - Wide Area Network
- VLAN - Virtual Local Area Network
- ACL - Access Control List
- VTP - VLAN Trucking Protocol
- HSRP - Hot Standby Router Protocol
- CCTV - Closed-Circuit Television
- L2 - Layer 2 (Data Link Layer)
- L3 - Layer 3 (Network Layer)
- Wi-Fi - Wireless Fidelity
- DHCP - Dynamic Host Configuration Protocol
- DNS - Domain Name System
- EIGRP - Enhanced Interior Gateway Routing Protocol
- IGP - Interior Gateway Protocol
- NAT - Network Address Translation
- AP - Access Point
- SYSLOG - System Logging Protocol
- SSH - Secure Shell
- PING - Packet Internet Grope

CHAPTER 1 - INTRODUCTION

1.1 Background of the Study

In the modern digital era, computer networks play a critical role in the efficient operation of organizations across all industries. The hospitality industry relies heavily on computer networking technologies to manage reservations, billing systems, staff coordination, security monitoring, and guest internet services. A well-designed network infrastructure ensures smooth communication, high availability, data security, and customer satisfaction.

Hotels today are no longer limited to basic communication needs. They require advanced networking solutions to support online booking platforms, property management systems (PMS), point-of-sale (POS) systems in restaurants and bars, surveillance systems, and high-speed wireless internet access for guests. Any failure or weakness in the network can directly affect hotel operations, service quality, and reputation.

This project focuses on designing and implementing a secure, scalable, and reliable computer network for **Golden Hotel**, addressing the limitations of its existing network infrastructure by applying modern networking concepts and best practices learned during the course.



Figure 1:Golden Hotel

1.2 Organization Overview – Golden Hotel

Golden Hotel is a medium-sized hotel consisting of **30 guest rooms distributed across three floors**. The hotel provides a range of facilities and services to its guests, including a **reception area, kitchen, restaurant, bar**, and a **swimming pool**. The hotel caters to both local and international guests and operates throughout the year.

The organizational structure of Golden Hotel includes several departments:

- Front Office / Reception
- Kitchen and Restaurant
- Bar Service
- Housekeeping
- Administration
- Finance and Management

Each department depends on continuous network access for day-to-day operations such as booking management, billing, inventory tracking, internal communication, and internet access. Additionally, the hotel provides wireless internet services to guests, which increases the demand for network security and performance.

Due to the increasing reliance on digital services, Golden Hotel requires a professionally designed network infrastructure that supports current needs while allowing for future expansion.

1.3 Existing Network Scenario

The existing network at Golden Hotel is based on a basic flat network architecture with minimal security and management features. All departments share the same network without proper segmentation. IP addresses are manually assigned, and there is no centralized server for network services.

Key characteristics of the existing network include:

- Single router for the entire hotel
- No VLAN implementation
- No redundancy or failover mechanisms

- Limited security controls
- Shared wireless network for staff and guests

This outdated setup creates several technical and operational challenges that affect both hotel staff and guests.

1.4 Problem Statement

The current network infrastructure of Golden Hotel suffers from poor performance, lack of security, and limited scalability. The absence of network segmentation exposes sensitive administrative systems to unauthorized access. The lack of redundancy means that a single device failure can disrupt all hotel operations.

Furthermore, the use of manual IP addressing increases the risk of IP conflicts and complicates troubleshooting. Guest and staff wireless networks are not separated, creating serious security concerns. There is also no proper monitoring or logging mechanism to identify and resolve network issues efficiently.

These issues highlight the need for a redesigned network solution that aligns with modern networking standards and organizational requirements.

1.5 Aim of the Project

The main aim of this project is to **design and implement a secure, efficient, and scalable computer network for Golden Hotel**, capable of supporting all departments and guest services while ensuring high availability and data security.

1.6 Objectives of the Project

The specific objectives of this project are:

- To analyze the existing network limitations at Golden Hotel
- To design a VLAN-based network for departmental separation
- To implement centralized IP address management using DHCP and VLSM
- To introduce redundancy using HSRP for uninterrupted network availability

- To enhance security using ACLs, SSH, port security, and NAT
- To design separate wireless networks for staff and guests
- To use virtualization (VMware) for efficient server deployment
- To test and validate the network design in a laboratory environment.

1.7 Scope of the Project

The scope of this project includes network design, simulation, configuration, testing, and documentation. The implementation is carried out in a **lab environment** using simulation and virtualization tools. Physical deployment at a real hotel site is not included.

The project covers:

- Logical and physical network design
- Network device configuration
- Security implementation
- Performance testing
- Documentation and reporting

1.8 Significance of the Project

This project demonstrates the practical application of computer networking theories learned during the academic program. It helps students gain hands-on experience in designing enterprise-level networks and prepares them for real-world networking roles.

For Golden Hotel, the proposed network design offers a blueprint for a reliable, secure, and future-ready network infrastructure that can improve operational efficiency and guest satisfaction.

1.9 Organization of the Report

This report is structured to provide a clear and systematic presentation of the project, from the background of the organization to the implementation and evaluation of the proposed network solution. The chapters are organized as follows:

- **Abstract** – Provides a brief introduction to the organization, the problems they face, and the solution proposed, written in simple, non-technical language.
- **Chapter 1: Introduction** – Introduces the organization, outlines the problem domain, describes the proposed solution, and states the aim and objectives of the project. It also gives an overview of the technologies and protocols implemented.
- **Chapter 2: Network Design** – Presents the detailed network design, including topology diagrams, IP address planning, rack and floor plans, Wi-Fi heat maps, bandwidth calculations, and port mappings. It also highlights the technologies and protocols used.
- **Chapter 3: Implementation** – Shows the actual implementation of the network, including screenshots of router, Layer 2 and Layer 3 switch configurations, VMware/DHCP setup, and real photos of the lab implementation.
- **Chapter 4: Evaluation** – Provides the results of testing and validation, including command outputs (show vlan, show ip nat translation), SSH logins, syslog server logs, and DHCP pool utilization.
- **Chapter 5: Conclusion / Conclusion & Future Work** – Summarizes the project outcomes, lessons learned, and recommendations for future enhancements.
- **References** – Lists all sources used in the report following the Harvard referencing style.
- **Annexure 1** – Details the specifications of all hardware devices used in the project.
- Summary of project outcomes
- Lessons learned
- Recommendations for future enhancements

This organization ensures a logical flow of information, guiding the reader from understanding the problem to seeing the practical implementation and evaluation of the network solution.

CHAPTER 2 – Network Design

2.1 Overview of Network Design

This section provides a summary of the network design approach, focusing on how the network will meet the organization's requirements for reliability, scalability, and security. The design ensures smooth communication between devices, efficient use of resources, and support for both current and future needs.

2.2 Network Topology Design

2.2.1 Physical Network Topology

Illustrates the actual layout of devices, including routers, switches, servers, and endpoints. This includes the physical connections between devices across floors and rooms.

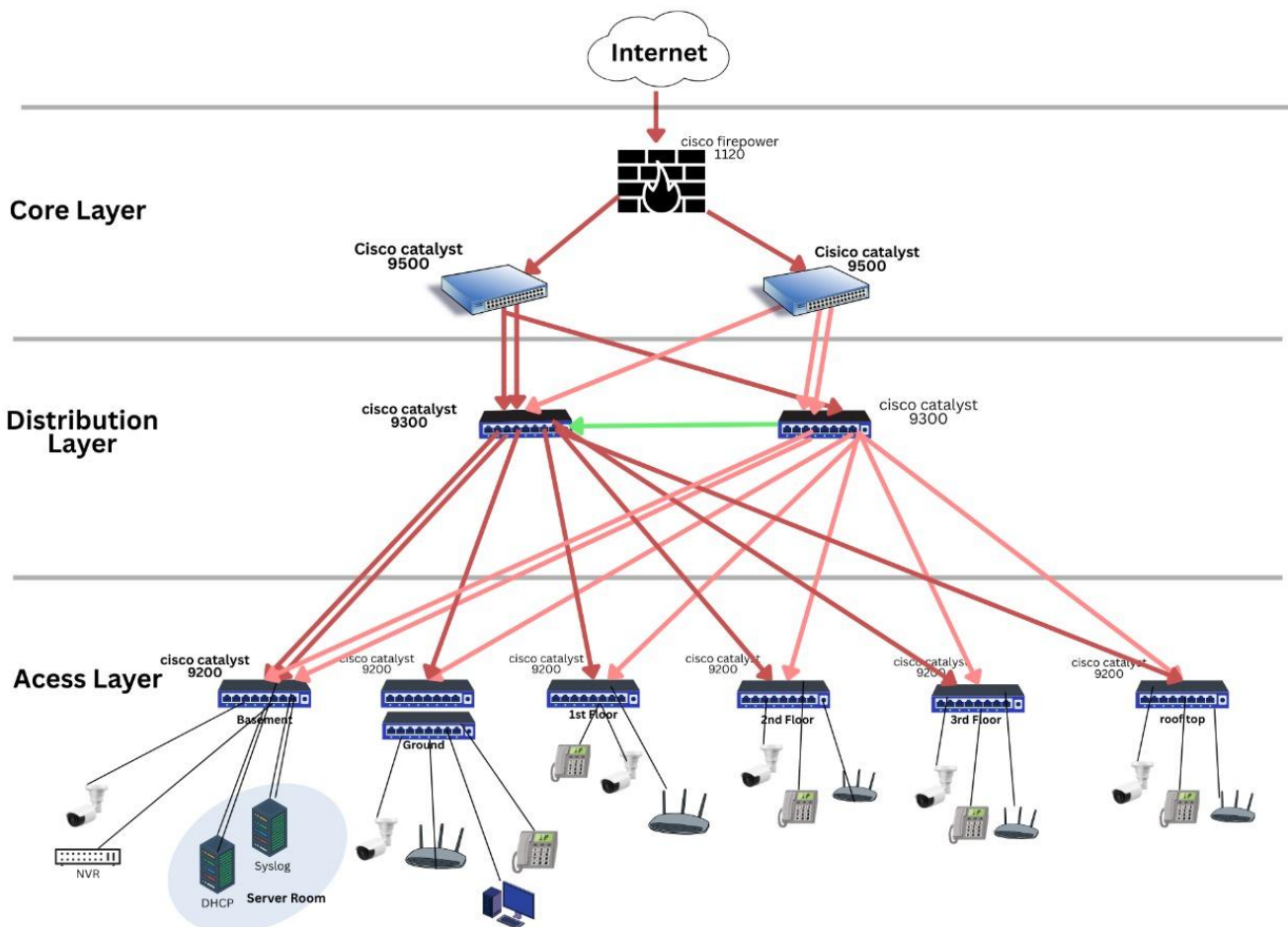


Figure 2: Physical Network Diagram for Golden Hotel

2.3 IP Addressing Design

2.3.1 IP Addressing Scheme

Explains the IP address allocation for all network devices, subnets, servers, and wireless devices to ensure efficient use of addresses and support future growth.

2.3.2 VLSM Design Table

Presents the Variable Length Subnet Mask (VLSM) table, showing subnet IDs, IP ranges, subnet masks, and the purpose of each subnet.

Table 1: Ip Address Table

VLAN ID	NAME	SUBNET and DEFAULT GATEWAY
99	For the Network Devices	192.168.0.0/24
10	Front Office	192.168.10.0/24
11	Guest Relations	192.168.11.0/24
12	Food & Beverage	192.168.12.0/24
13	HR	192.168.13.0/24
14	Finance	192.168.14.0/24
15	IT	192.168.15.0/24
20	Staff(General/personal use)	192.168.21.0/22
30	Guest Wifi	192.168.30.0/22
40	CCTV (NVR)	192.168.40.2
50	IP phone	192.168.50.0

2.4 Rack Layout and Equipment Placement

Describes how networking equipment is organized within racks for optimal accessibility, airflow, and cable management.

2.4.1 Network Equipment Rack on the Basement

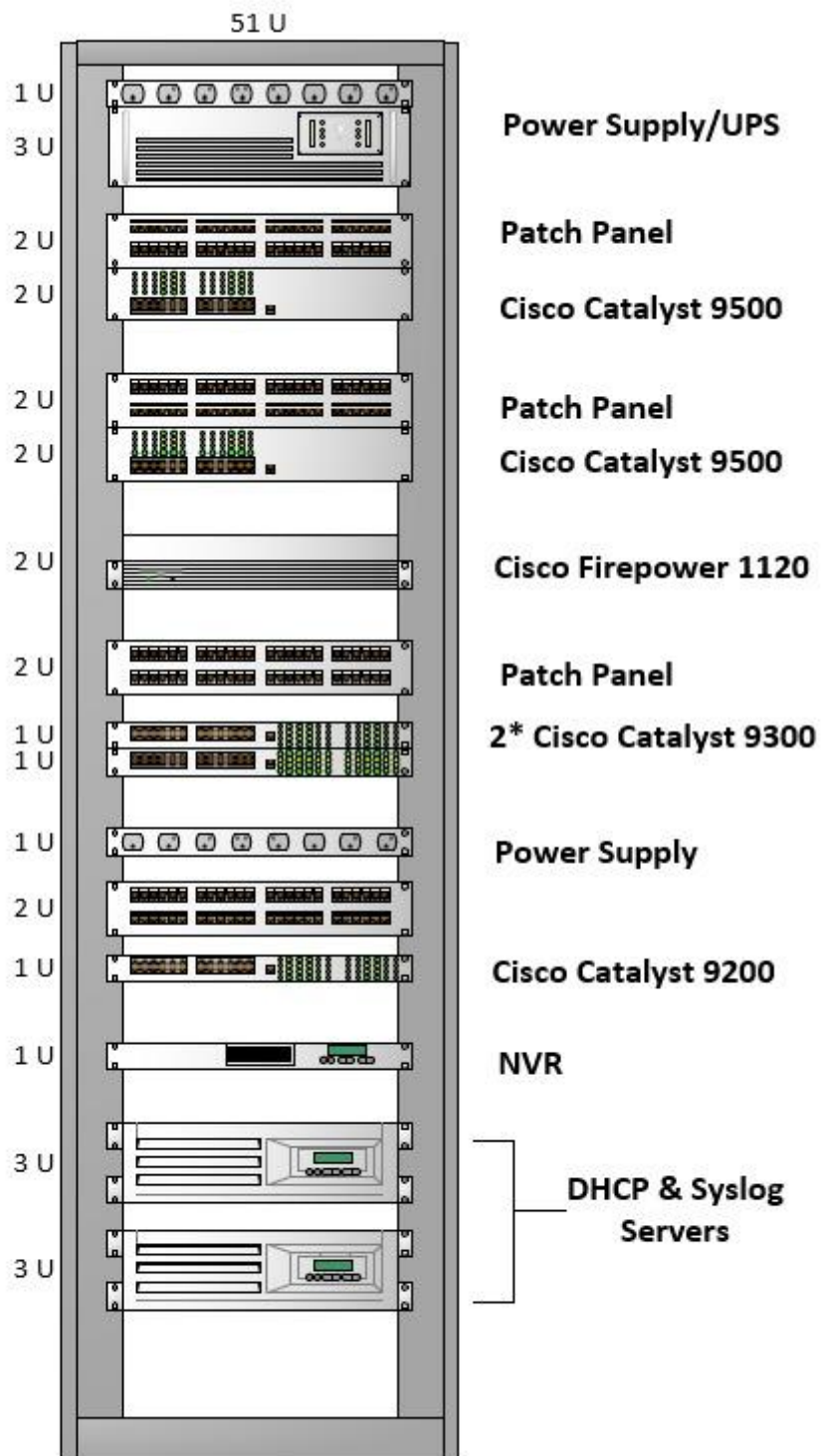


Figure 3: Network Rack Diagram of Basement

2.4.2 Network Equipment Rack on the Ground Floor

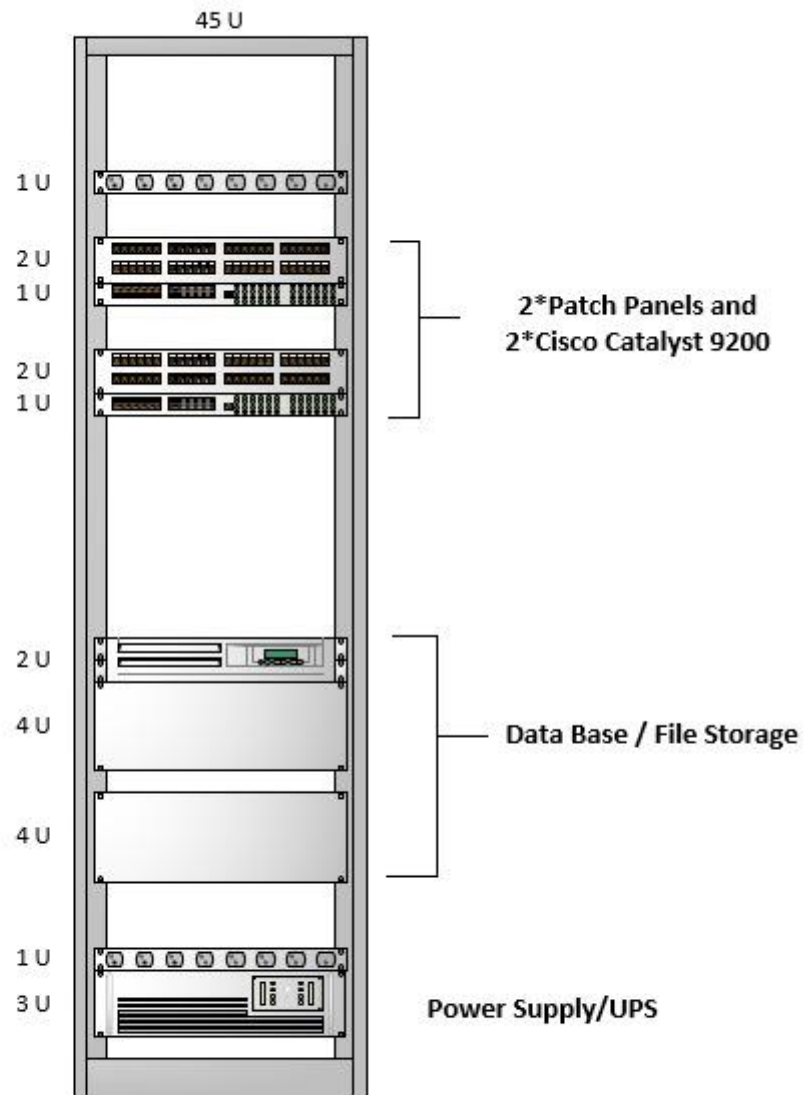


Figure 4: Network Rack Diagram of Ground floor

2.4.3 Network Equipment Rack on 2,3,4 Floor

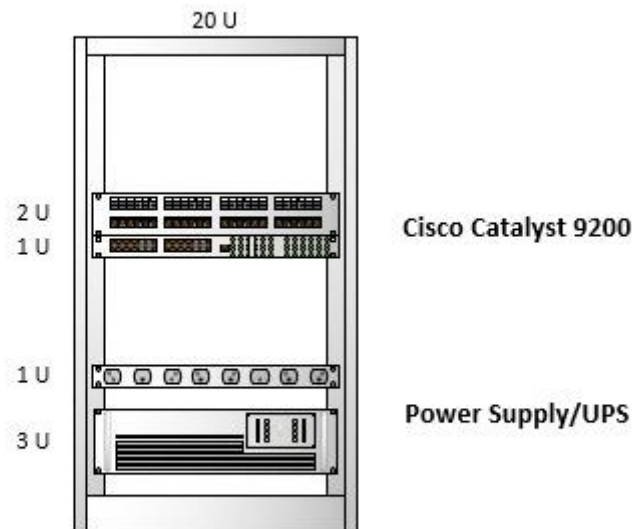


Figure 5: Network Rack Diagram of 2,3,4 floor

2.5 Floor Plan Network Design

Shows the placement of network devices and cabling across different floors and areas.

Highlights server rooms, workstations, and network closets.

2.5.1 Ground Floor Design

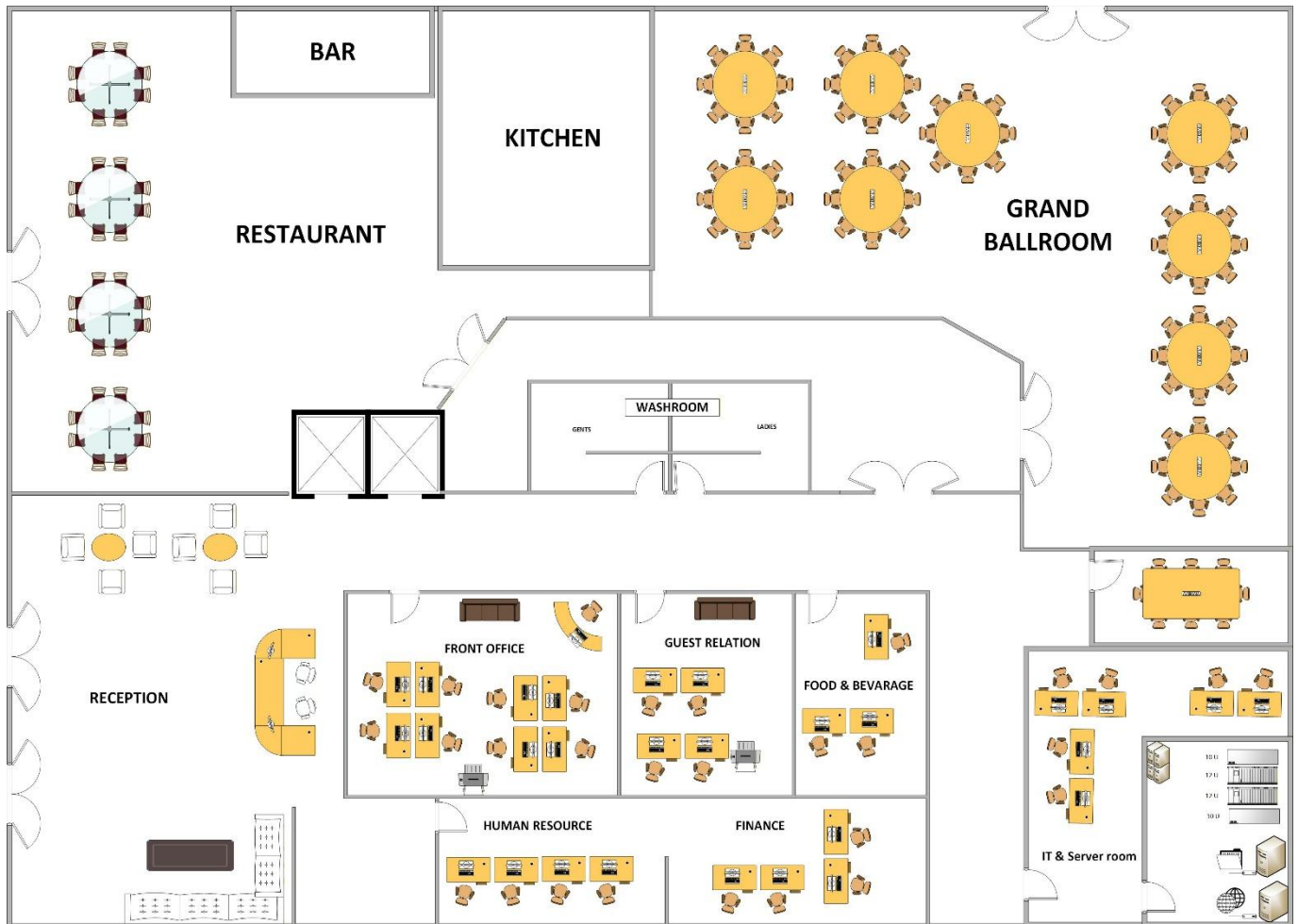


Figure 6: Ground Floor design plan of the Golden Hotel

2.5.2 1st,2nd and 3rd Floor Design

(All three of these floors mentioned have the same design)



Figure 7: 1st,2nd & 3rd floor design plan of the Golden Hotel

2.5.3 4th Floor Design (Rooftop)

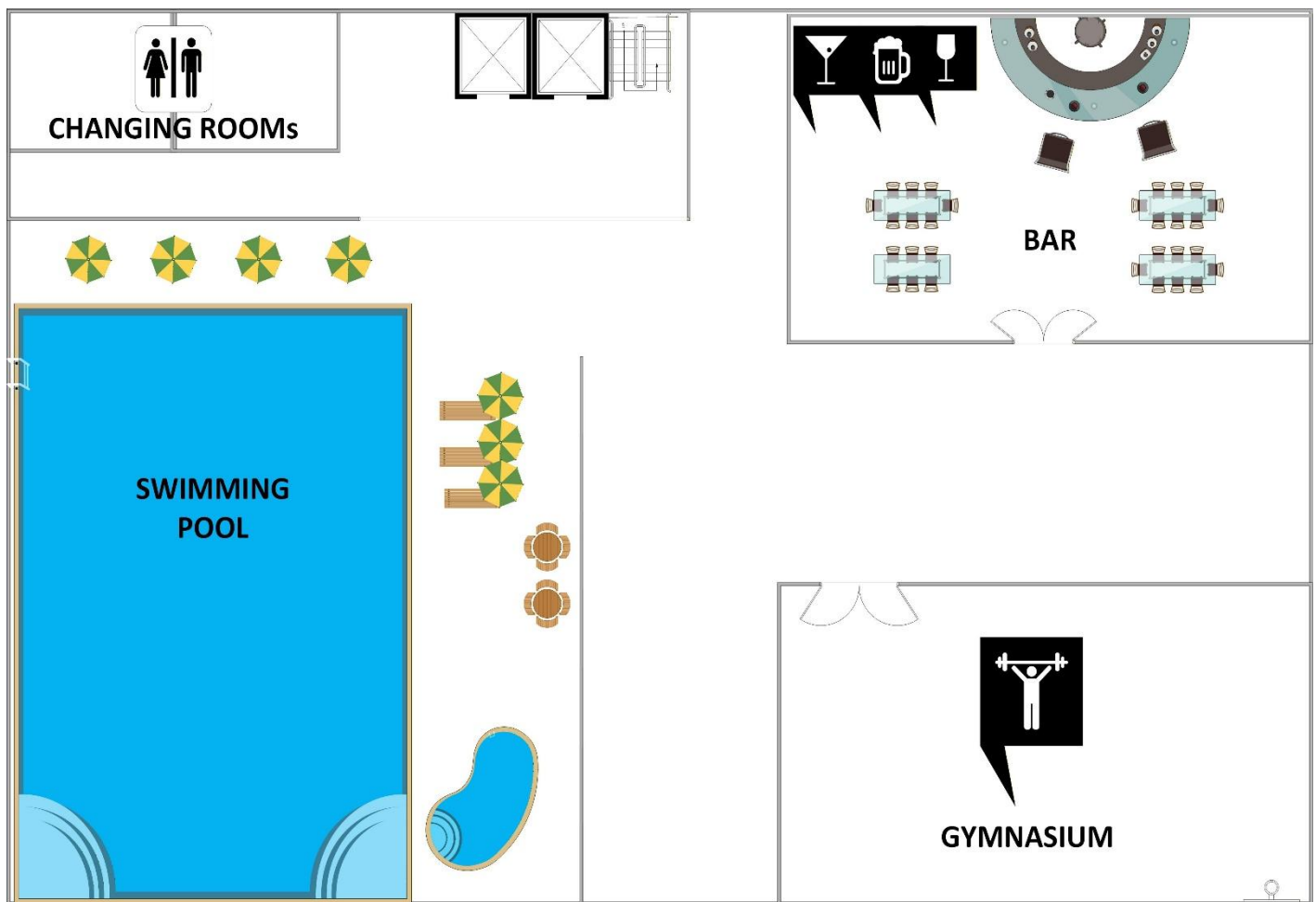


Figure 8: 4th floor (Rooftop) design plan of the Golden Hotel

3.5.4 Basement floor

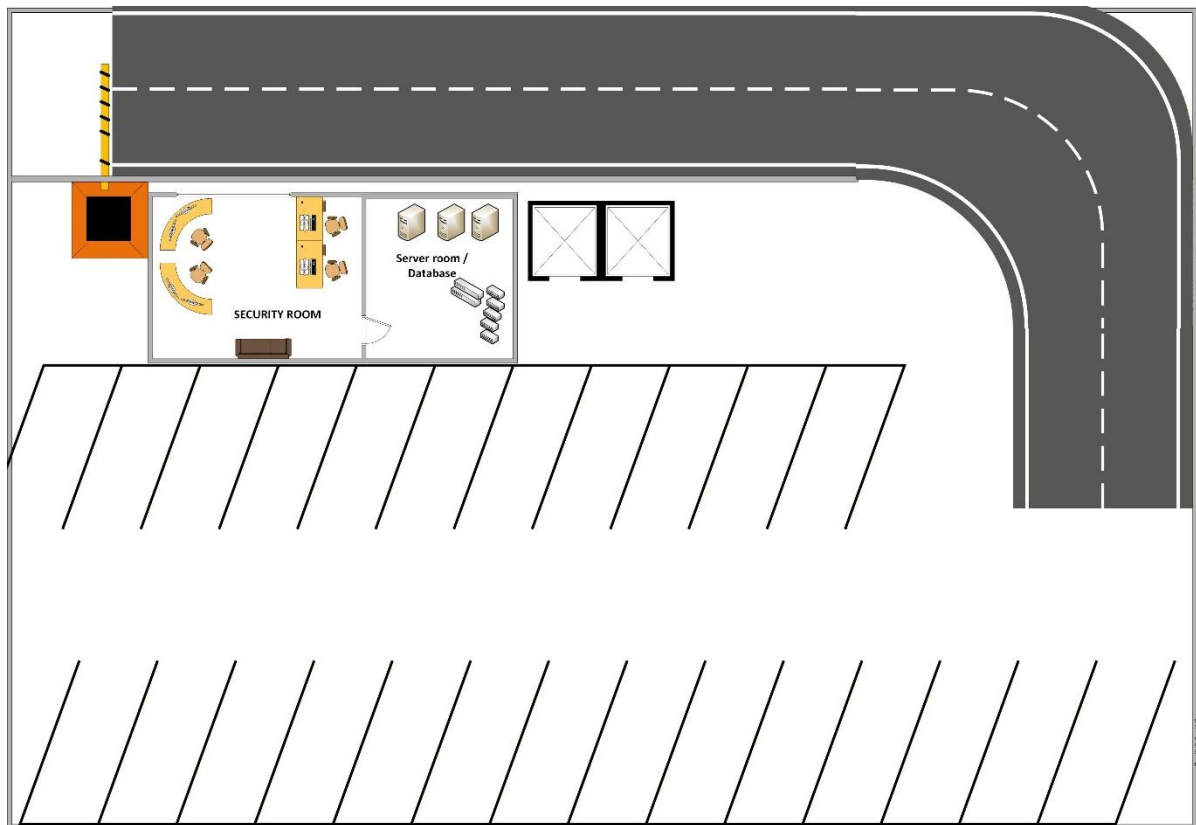


Figure 9: Basement design plan of the Golden Hotel

2.6 Wireless Network Design

2.6.1 Access Point Placement

Describes the planned locations of wireless access points to ensure full coverage, capacity, and minimal interference.

2.6.2 Wi-Fi Coverage and Heat Map

Presents Wi-Fi coverage using heat maps, indicating signal strength, dead zones, and areas with overlapping coverage.

2.6.2.1 Ground Floor heat map

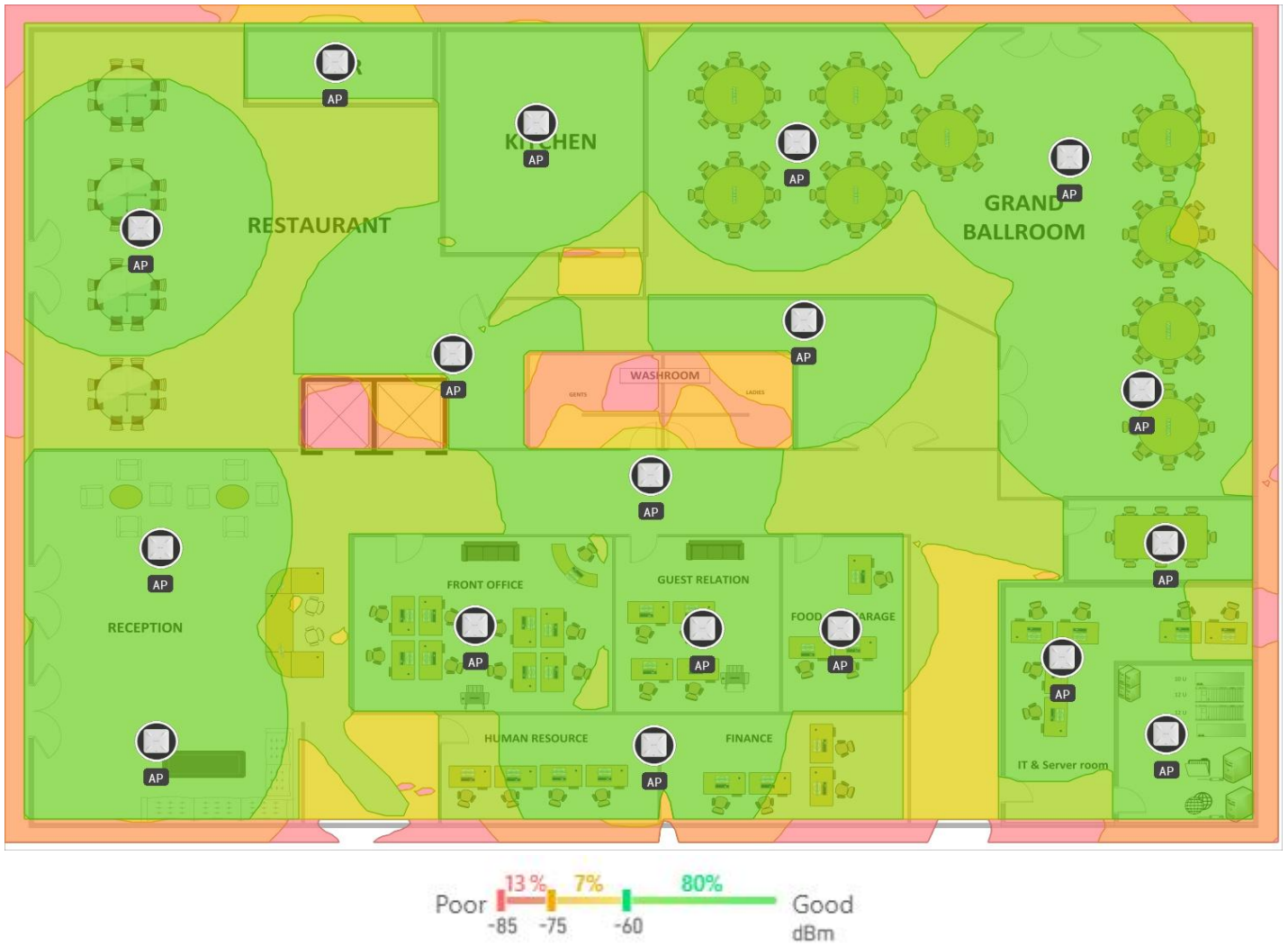


Figure 10: Wi-Fi heat map of ground floor golden hotel

2.6.2.2 1st, 2nd and 3rd Floor Wi-Fi heat maps



Figure 11: Wi-Fi heat map of ground floor golden hotel

2.6.2.3 4th Floor Design (Rooftop)



Figure 12: Wi-Fi heat map of 4th floor (rooftop) golden hotel

2.6.2.4 Basement Heat Map design

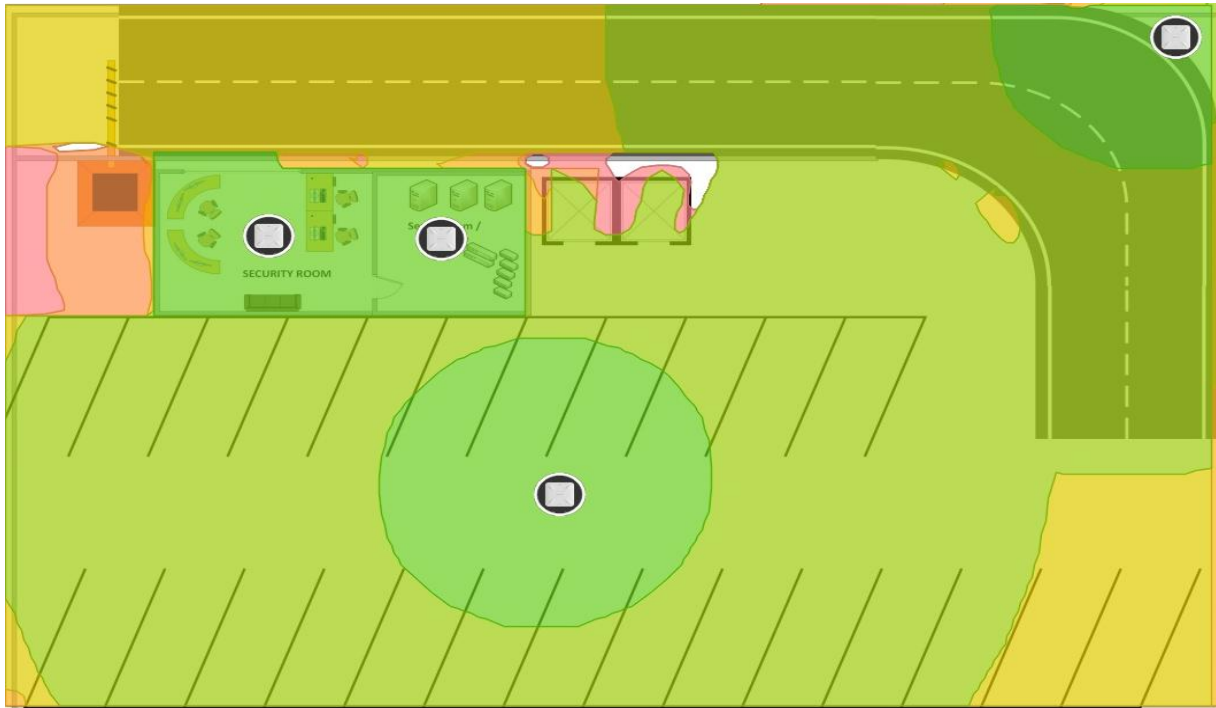


Figure 13: Wi-Fi heat map of Basement golden hotel.

2.7 Technologies Used and Implemented Protocols

Lists all technologies and protocols implemented in the network design, including:

- Routing protocols: OSPF/EIGRP
- VLAN segmentation for logical separation
- HSRP/VRRP for redundancy
- DHCP for automated IP address management
- ACLs and security measures
- Wireless standards (Wi-Fi 6, WPA3)

2.8 Summary of Network Design

Summarizes the key points of the network design, emphasizing how the topology, IP planning, wireless coverage, bandwidth calculations, and port mapping work together to create a secure, efficient, and scalable network.

2.9 Bandwidth calculation

Design Assumptions

- Golden Key Hotel has 30 guest rooms
- Each room may connect 2 devices on average
- Maximum allocated bandwidth per room: 30 Mbps (shared)
- Not all rooms use full bandwidth at the same time
- Peak usage factor considered: 50% concurrency
- Separate bandwidth allocated for hotel staff and services

Peak Room Usage Calculation

Total rooms = 30

Peak usage factor = 50%

Active rooms at peak = $30 \times 0.5 = 15$ room

Guest Bandwidth Requirement

Bandwidth per room = 30 Mbps

Guest bandwidth = 15×30 Mbps

= 450 Mbps

Total Bandwidth Requirement

Total bandwidth = Guest bandwidth + Staff bandwidth

= $450 + 110$

= 560 Mbps

Safety and Future Growth Margin (25%)

Final bandwidth = 560×1.25

= 700 Mbps

staff and Operational Bandwidth

Area	Estimated Bandwidth
Reception & Front Office	30 Mbps
Administration & Finance	20 Mbps
Restaurant & Bar (POS systems)	30 Mbps
Kitchen systems	10 Mbps
Management	20 Mbps
Total Staff Bandwidth	110 Mbps

2.10 Bandwidth Calculation

Bandwidth calculation was performed to ensure uninterrupted network services for guests and hotel operations. Golden Key Hotel consists of 30 guest rooms; each allocated a maximum shared bandwidth of 30 Mbps. Considering a peak usage factor of 50%, approximately 15 rooms are expected to be actively using the network during peak hours.

The estimated guest bandwidth requirement is calculated as 450 Mbps. Additional bandwidth was allocated for hotel staff operations including reception, administration, restaurant, bar, kitchen, and management systems, totaling approximately 110 Mbps.

The combined bandwidth requirement is approximately 560 Mbps. A safety margin of 25% was added to accommodate future expansion and traffic fluctuations, resulting in a final bandwidth requirement of approximately 700 Mbps. Therefore, a 1 Gbps fiber internet connection is recommended for the hotel network.

CHAPTER 3 – IMPLEMENTATION

3.1 Introduction

This chapter describes the practical implementation of the proposed network design for Golden Hotel. The implementation was carried out in a **laboratory environment** using networking devices and virtualization platforms. The objective of this phase is to configure, test, and validate the designed network according to the requirements identified earlier.

The implementation includes router configuration, Layer 3 and Layer 2 switch configuration, VLAN creation, redundancy configuration, security implementation, DHCP services using VMware, and wireless network setup.

3.2 Implementation Environment

The network implementation was performed using the following environment:

- Cisco routers and switches (physical or simulated)
- VMware Workstation for server virtualization
- Client devices for testing
- Console and SSH access for configuration

All configurations were tested to ensure correct connectivity, security, and redundancy.

3.3 Router Configuration

3.3.1 Basic Router Configuration

Each router was configured with:

- Hostname
- Password encryption
- Console and VTY passwords
- Secure Shell (SSH) access
- IP addressing

```
Router>EN
Router#CONF T
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#ho
Router(config)#hostname R1
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#int
R1(config)#interface gig
R1(config)#interface gigabitEthernet 0/0.10
R1(config-subif)#en
R1(config-subif)#encapsulation dot
R1(config-subif)#encapsulation dot1Q
% Incomplete command.
```

Figure 14: host name configuration

```

R1(config)#
R1(config)#
R1(config)#int
R1(config)#interface gig
R1(config)#interface gigabitEthernet 0/0.10
R1(config-subif)#en
R1(config-subif)#encapsulation dot
R1(config-subif)#encapsulation dot1Q
% Incomplete command.

R1(config-subif)#interface gigabitEthernet 0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip
R1(config-subif)#ip add
R1(config-subif)#ip ad
R1(config-subif)#ip add
R1(config-subif)#ip address 192.168.10.254 255.255.255.0
R1(config-subif)#
R1(config-subif)#
R1(config-subif)#interface gigabitEthernet 0/0.13
R1(config-subif)#encapsulation dot1Q 13
R1(config-subif)#ip address 192.168.13.254 255.255.255.0
R1(config-subif)#
R1(config-subif)#
R1(config-subif)#
R1(config-subif)#interface gigabitEthernet 0/0.15
R1(config-subif)#encapsulation dot1Q 15
R1(config-subif)#ip address 192.168.15.254 255.255.255.0
R1(config-subif)#
R1(config-subif)#
R1(config-subif)#exit
R1(config)#
R1(config)#
R1(config)#int
R1(config)#interface gig
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#no shut
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#
R1(config-if)#
*Dec 16 23:37:08.655: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to downe
*Dec 16 23:37:11.567: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Dec 16 23:37:12.567: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
% Ambiguous command: "a"

```

Figure 15: IP addressing on router

SSH Configuration on R1

```

Router(config)#hostname R1
R1(config)#crypto key generate rsa modulus 2048
The name for the keys will be: R1.golden.local

% The key modulus size is 2048 bits
% Generating 2048 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 6 seconds)

R1(config)#ip ssh version 2
R1(config)#ip ssh ver
R1(config)#ip ssh version 2
R1(config)#
R1(config)#line
R1(config)#line vt
R1(config)#line vty 0 15
R1(config-line)#log
R1(config-line)#login
R1(config-line)#login lo
R1(config-line)#login local
R1(config-line)#tran
R1(config-line)#transport in
R1(config-line)#transport ssh
R1(config-line)#transport in
R1(config-line)#transport input ss
R1(config-line)#transport input ssh
R1(config-line)#end
R1#sh ip ssh
SSH Enabled - version 2.0
Authentication methods:publickey,keyboard-interactive,password
Encryption Algorithms:aes128_cbc_aes192_cbc_aes256_cbc_aes128_cbc_3des_cbc_aes192_cbc_aes256_cbc

```

Figure 16: configuring the SSH

```

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip ssh version 2
!

```

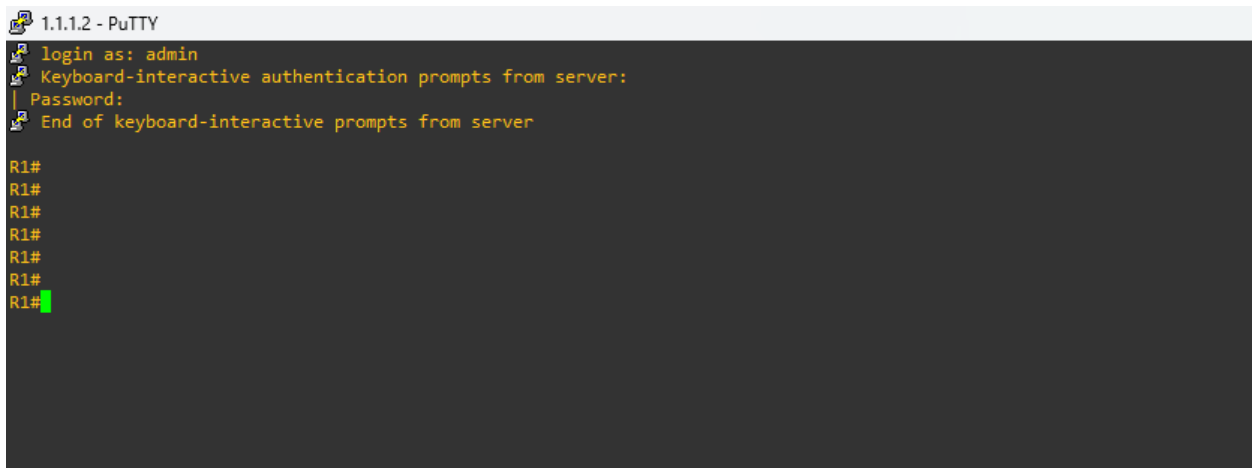
Figure 17: configuring the SSH 2

```

!
line con 0
  stopbits 1
line aux 0
  stopbits 1
line vty 0 4
  login local
  transport input ssh
line vty 5 15
  login local
  transport input ssh
!

```

Figure 18: SSH Configuration



```
1.1.1.2 - PuTTY
login as: admin
Keyboard-interactive authentication prompts from server:
Password:
End of keyboard-interactive prompts from server

R1#
R1#
R1#
R1#
R1#
R1#
R1#
```

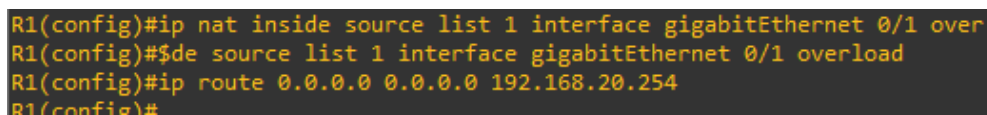
Figure 19: SSH login to R1

3.3.2 Network Address Translation (NAT)

NAT was configured to allow internal private IP addresses to access external networks. This improves security by hiding internal addressing and enables internet access for hotel departments.

Types of NAT implemented:

- Static NAT (if required)
- Dynamic NAT or PAT for internet access



```
R1(config)#ip nat inside source list 1 interface gigabitEthernet 0/1 over
R1(config)#$de source list 1 interface gigabitEthernet 0/1 overload
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.20.254
R1(config)#
```

Figure 20: NAT configuration

```

Enter configuration commands, one per line. End with CNTL/Z.
L3SW(config)#
L3SW(config)#
L3SW(config)#
L3SW(config)#
L3SW(config)#ip rout
L3SW(config)#ip rout
L3SW(config)#ip rout
L3SW(config)#ip route
L3SW(config)#ip route
L3SW(config)#ip route ?
  A.B.C.D  Destination prefix
  profile  Enable IP routing table profile
  static   Allow static routes
  topology Configure static route for a Topology Routing/Forwarding instance
  vrf      Configure static route for a VPN Routing/Forwarding instance
L3SW(config)#ip route 0.0.0.0
L3SW(config)#ip route 0.0.0.0 ?
  A.B.C.D  Destination prefix mask
L3SW(config)#ip route 0.0.0.0 0.0.0.0 ?
  A.B.C.D      Forwarding router's address
  ANI          Autonomic-Networking virtual interface
  Auto-Template Auto-Template interface
  CEM-PG       Circuit Emulation interface with Protection group
  Capwap       Capwap tunnel interface
  DHCP         Default Gateway obtained from DHCP
  GMPLS        MPLS interface
  GigabitEthernet GigabitEthernet IEEE 802.3z
  InternalInterface Internal Interface
  LISP         Locator/ID Separation Protocol Virtual Interface
  Loopback     Loopback interface
  Null         Null interface
  PROTECTION_GROUP Protection-group controller
  Port-channel Ethernet Channel of interfaces
  Tunnel       Tunnel interface
  Tunnel-tp    MPLS Transport Profile interface
  Vlan         Catalyst Vlans
L3SW(config)#ip route 0.0.0.0 0.0.0.0 1.1.1.2
L3SW(config)#do ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 95/107/112 ms
L3SW(config)#admin

```

Figure 21: confirmation of NAT configuration

3.4 Layer 3 Switch Configuration

3.4.1 VLAN Configuration

VLANs were created for each department in Golden Hotel, such as:

- HR
- Front Office
- Guest Relations
- Food & Beverage
- Finance
- IT
- Guest Wi-Fi
- Staff Wi-Fi
- CCTV
- IP Phones

Each VLAN was assigned a unique VLAN ID and IP subnet.

```
L3SW>
L3SW>
L3SW>
L3SW>
L3SW>en
L3SW#
L3SW#
L3SW#conf t
Enter configuration commands, one per line. End with CNTL/Z.
L3SW(config)#
L3SW(config)#
L3SW(config)#
L3SW(config)#
L3SW(config)#vlan 10
L3SW(config-vlan)#vlan 13
L3SW(config-vlan)#vlan 15
L3SW(config-vlan)#vlan 97
L3SW(config-vlan)#vlan 98
L3SW(config-vlan)#
L3SW(config-vlan)#
L3SW(config-vlan)#
L3SW(config-vlan)#exit
L3SW(config)#
L3SW(config)#exit
L3SW#
```

Figure 22: VLAN configuration in L3 switch

```

!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
  switchport mode trunk
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
  no ip address
  shutdown
!
interface Vlan10
  description ## Front_Office  ##
  ip address 192.168.10.254 255.255.255.0
!
interface Vlan13
  description ## HR  ##
  ip address 192.168.13.254 255.255.255.0
!
interface Vlan15
  description ## ACL - IT  ##
  ip address 192.168.15.254 255.255.255.0
!
interface Vlan97
  description ## Server  ##
  ip address 192.168.97.254 255.255.255.0
!
interface Vlan98
  description ## Up-Link  ##
  ip address 10.10.10.1 255.255.255.0
!
ip forward-protocol nd

```

Figure 23: VLAN configuration in L3 switch

```
!  
interface Vlan10  
  description ## Front_Office ##  
  ip address 192.168.10.254 255.255.255.0  
  ip helper-address 192.168.97.2  
!  
interface Vlan13  
  description ## HR ##  
  ip address 192.168.13.254 255.255.255.0  
  ip helper-address 192.168.97.2  
!  
interface Vlan15  
  description ## ACL - IT ##  
  ip address 192.168.15.254 255.255.255.0  
  ip helper-address 192.168.97.2  
!  
interface Vlan60  
  description ##Sales##  
  ip address 192.168.60.254 255.255.255.0  
  ip access-group 100 in  
!  
interface Vlan70  
  description ## F&B ##  
  ip address 192.168.70.254 255.255.255.0  
!  
interface Vlan97  
  description ## Server ##  
  ip address 192.168.97.254 255.255.255.0  
!  
!
```

Figure 24: VLAN configuration in L3 switch

3.4.2 Inter-VLAN Routing

Inter-VLAN routing was configured on the Layer 3 switch using Switch Virtual Interfaces (SVIs). This allows communication between departments while maintaining logical separation.

3.4.3 EtherChannel (Port Channel) Implementation

- Two switches connected with **multiple physical links**
- EtherChannel used to:
 - Increase bandwidth
 - Provide redundancy
 - Prevent STP blocking

•

```
config)# interface range gigabitEthernet0/1 - 2
config-if-range)# switchport mode trunk
config-if-range)# channel-group 1 mode active
config-if-range)# exit

config)# interface port-channel 1
config-if)# switchport mode trunk
config-if)# exit

config)# interface port-channel 1
config-if)# switchport mode trunk
config-if)# exit
```

Figure 25: EtherChannel Configuration

- EtherChannel Verification Commands
- Command to Show Summary
- show EtherChannel summary

3.4.4 HSRP Configuration (Primary & Backup)

To ensure high availability, **HSRP** was implemented between the primary and backup Layer 3 switches. If the primary switch fails, the backup switch automatically takes over.

Advantages:

- No single point of failure
- High availability
- Seamless failover

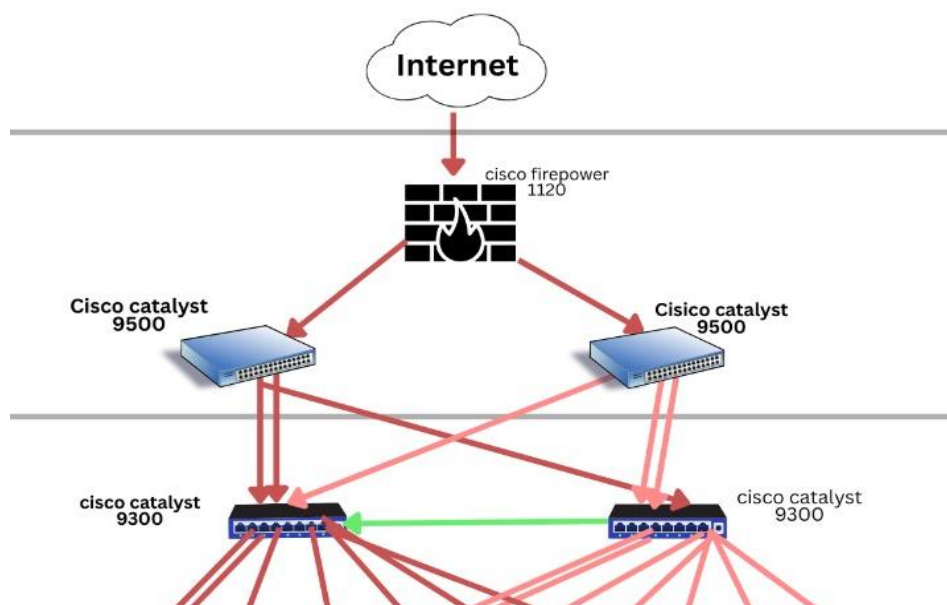


Figure 26: HSRP enabled in core layer and the distribution layer

3.4.5 Access Control List (ACL) Configuration

This section describes the implementation of Access Control Lists (ACLs) to restrict unauthorized traffic between VLANs and protect sensitive hotel departments.

```
access-list 100 deny ip 192.168.60.0 0.0.0.255 192.168.70.0 0.0.0.255
access-list 100 permit ip any any
```

Figure 27: ACL

3.5 Layer 2 Switch Configuration

Layer 2 switches were configured to connect end devices such as PCs, printers, and access points.

3.5.1 Access and Trunk Ports

- Access ports were assigned to specific VLANs
- Trunk ports were configured between switches to carry multiple VLANs

```
interface GigabitEthernet1/0/13
 switchport mode trunk
 channel-group 1 mode desirable
!
interface GigabitEthernet1/0/14
 switchport mode trunk
 channel-group 1 mode desirable
```

Figure 28: Trunk port & EtherChannel configuration

3.6 DHCP Server Implementation Using VMware

A virtual machine was created using **VMware Workstation** to act as a DHCP server. This server automatically assigns IP addresses to devices in different VLANs.

Features:

- Separate DHCP pools for each VLAN

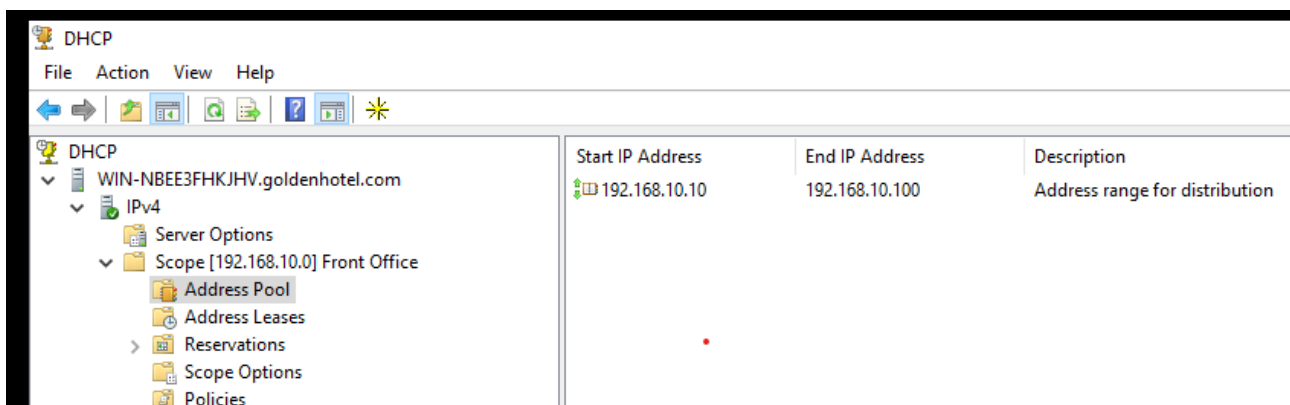


Figure 29: DHCP pools configured on the virtual server machine

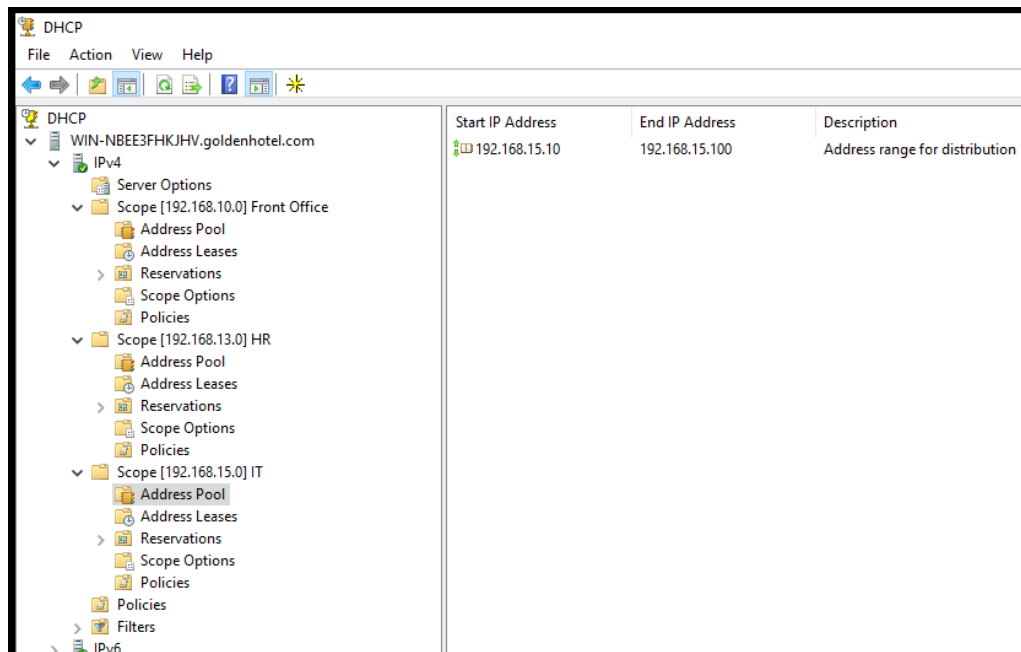


Figure 30: DHCP pools configured on the virtual server machine

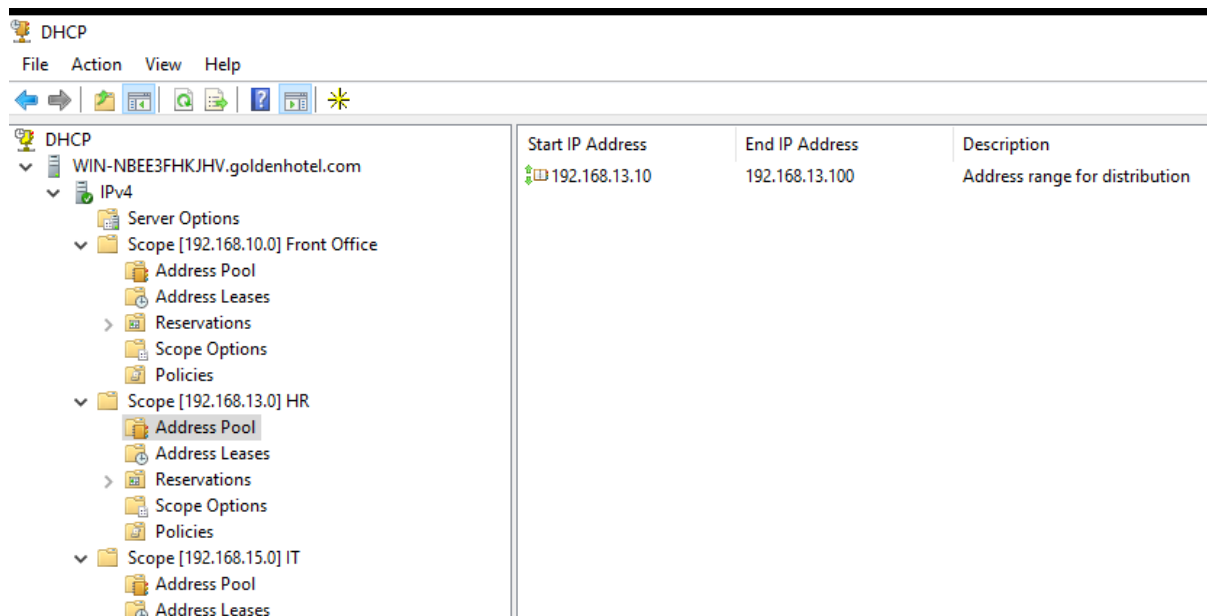


Figure 31: DHCP pools configured on the virtual server machine

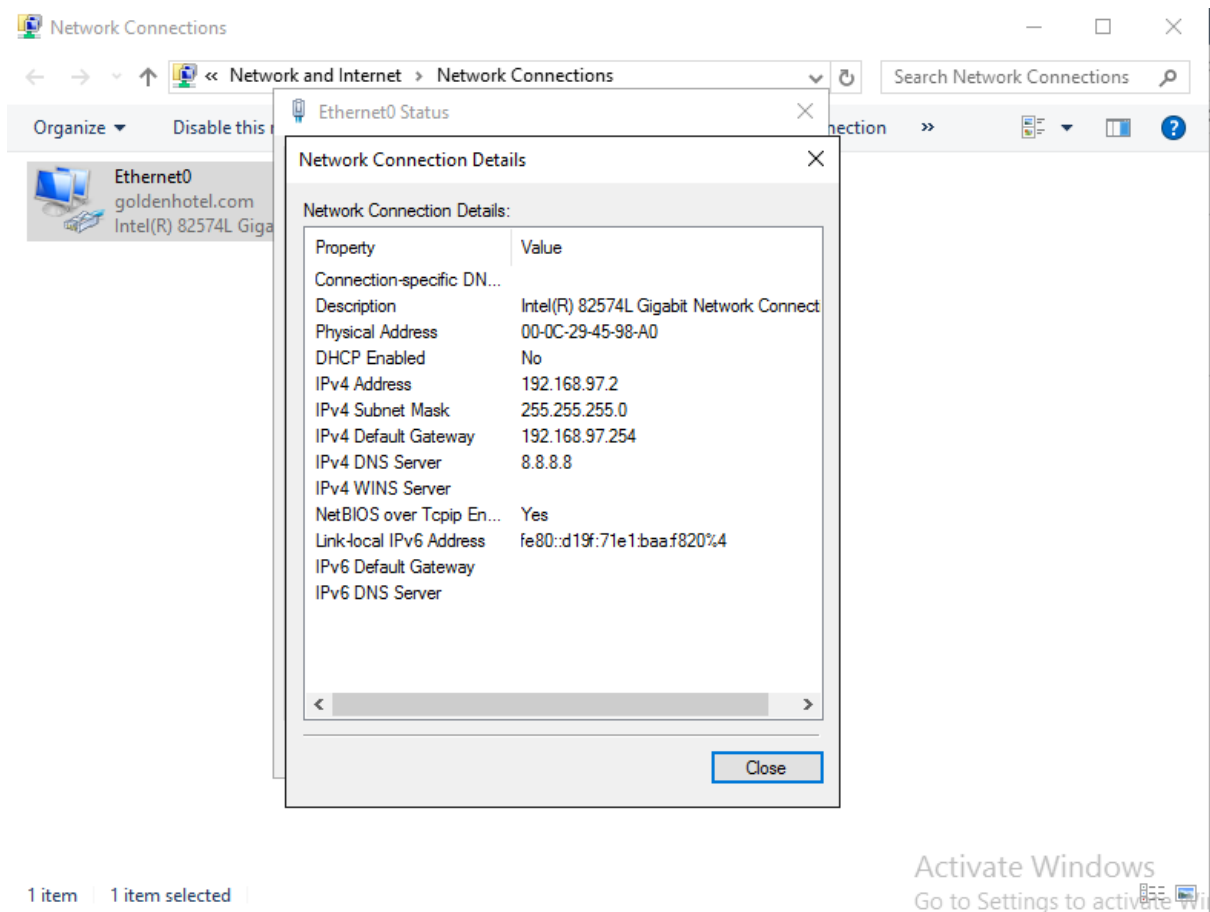


Figure 32: DHCP Ip Addresses are assigned to the client machines accordingly.

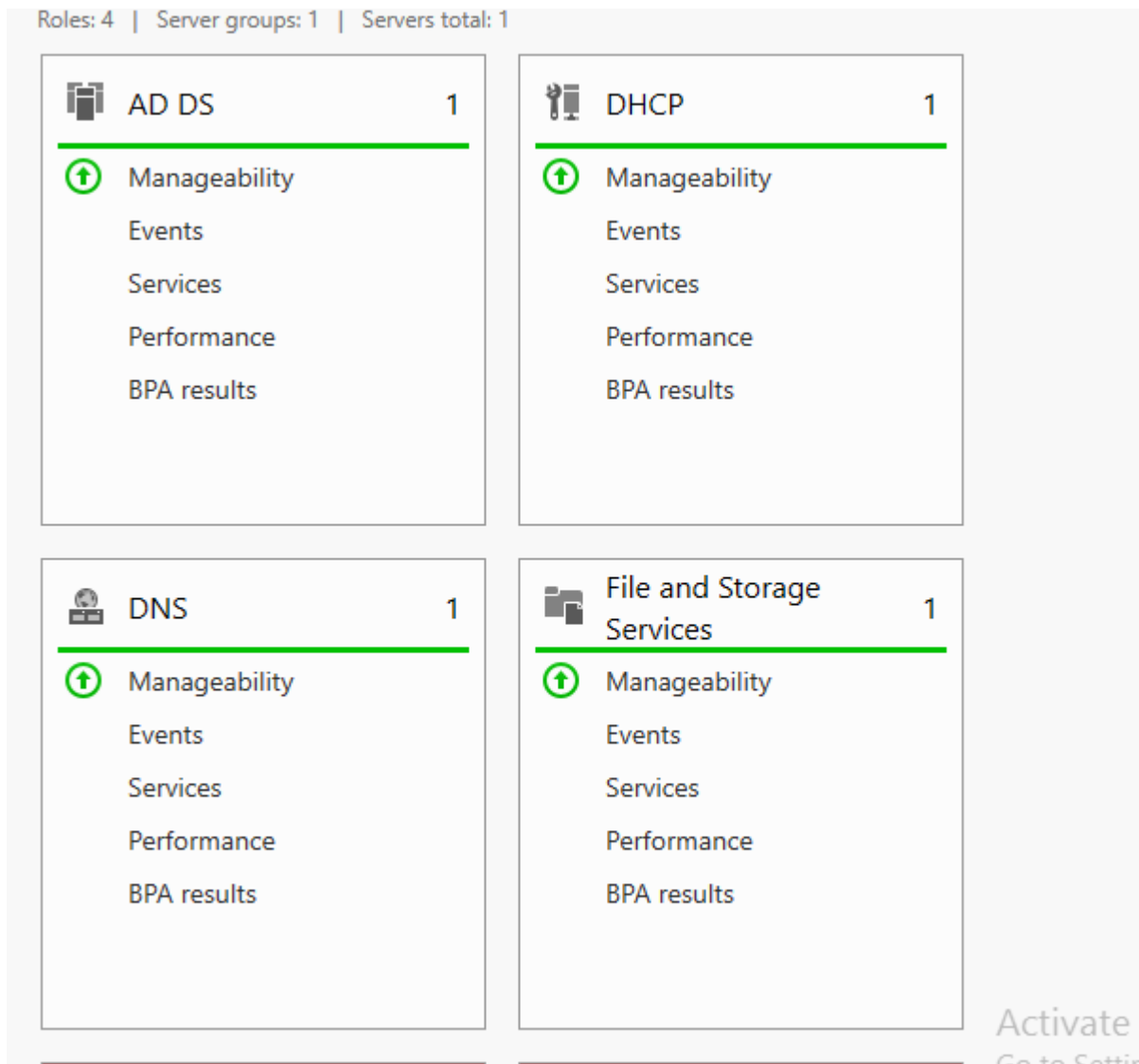


Figure 33: Windows Validation Server

3.7 Wireless Network Implementation

Wireless access points were configured to provide Wi-Fi coverage across all floors of the hotel.

- Separate SSIDs for staff and guests
- Guest network isolated from internal systems
- Secure encryption enabled

3.8 Syslog and Remote Management

A Syslog server was configured to collect logs logins through SSH in to the network devices. SSH was used for secure remote management.

Benefits:

- Centralized logging
- Improved troubleshooting
- Secure device access

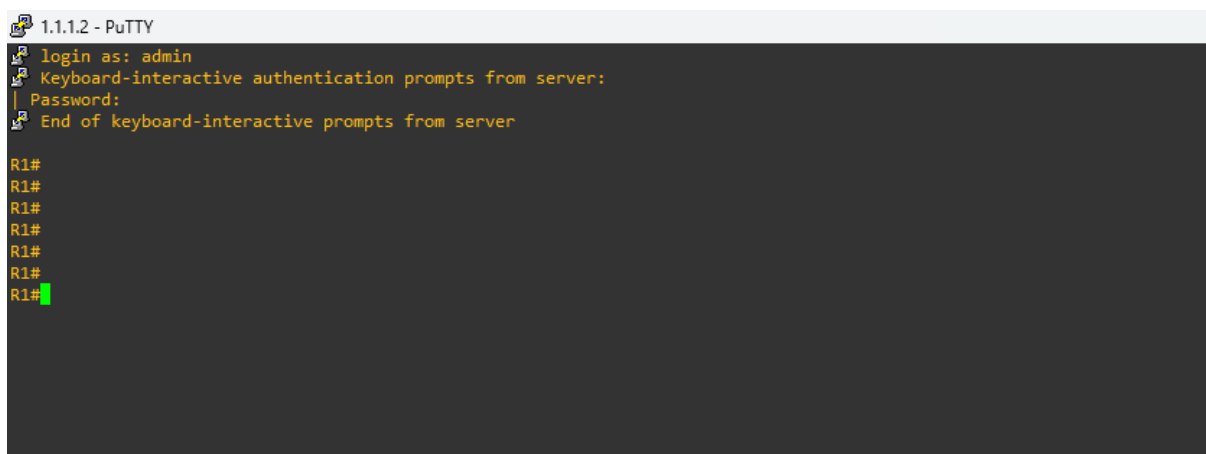


Figure 34: SSH login

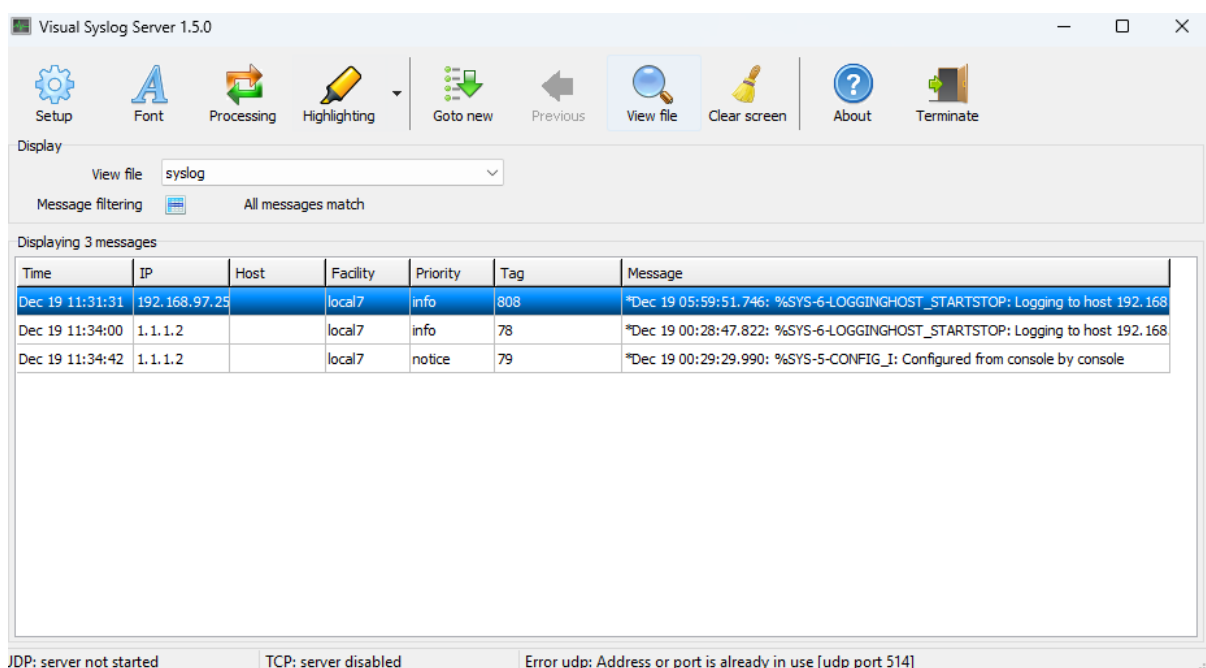


Figure 35: Syslog server Logs

3.9 Summary

This chapter explained the complete implementation process of the Golden Hotel network. All major network components were configured and tested successfully. The implemented network meets the requirements of security, availability, scalability, and performance.

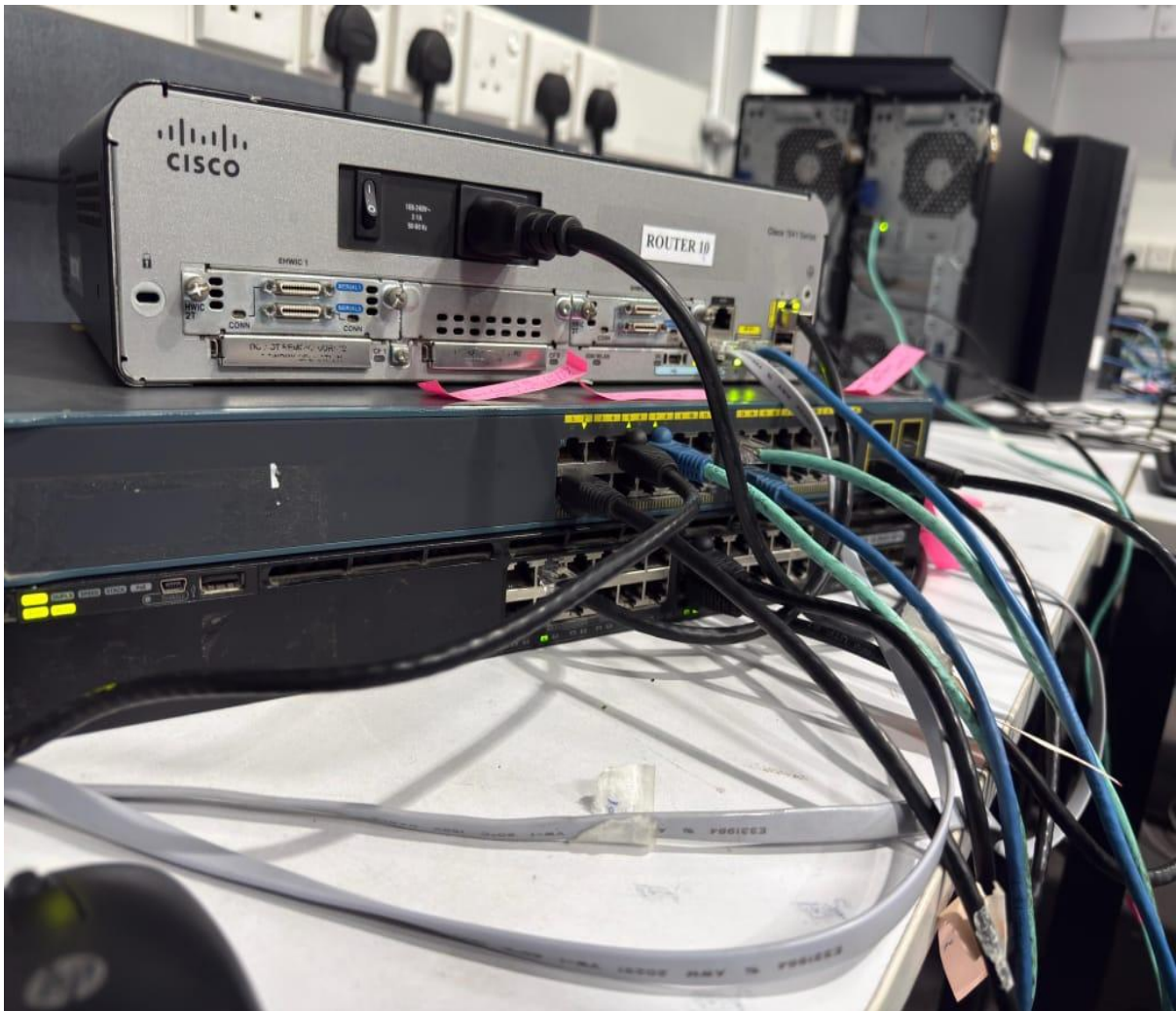


Figure 36: physical Implementation at the network lab



Figure 37: physical Implementation at the network lab

CHAPTER 4 – EVALUATION

4.1 Introduction

Evaluation is a critical phase of the network implementation process, as it verifies whether the designed and implemented network meets the specified objectives and requirements. This chapter focuses on validating the functionality, performance, security, and reliability of the Golden Hotel network through systematic testing and analysis.

The evaluation process includes testing VLAN configurations, routing and Network Address Translation (NAT), secure remote access using SSH, centralized logging using a Syslog server, and Dynamic Host Configuration Protocol (DHCP) pool utilization. Each test is conducted using standard network verification commands and tools to ensure accurate and reliable results.

4.2 VLAN Configuration Verification

4.2.1 Purpose of the Test

VLAN verification is performed to confirm that:

- All VLANs are created correctly
- Devices are assigned to the correct VLANs
- VLAN segmentation is functioning as designed

This ensures logical separation between departments such as reception, restaurant, administration, and guest networks.

4.2.2 Test Method

The following command is executed on the Layer 3 and Layer 2 switches:

```
!
interface Vlan10
description ## Front_Office  ##
ip address 192.168.10.254 255.255.255.0
ip helper-address 192.168.97.2
!
interface Vlan13
description ## HR  ##
ip address 192.168.13.254 255.255.255.0
ip helper-address 192.168.97.2
!
interface Vlan15
description ## ACL - IT ##
ip address 192.168.15.254 255.255.255.0
ip helper-address 192.168.97.2
!
interface Vlan60
description ##Sales##
ip address 192.168.60.254 255.255.255.0
ip access-group 100 in
!
interface Vlan70
description ## F&B ##
ip address 192.168.70.254 255.255.255.0
!
interface Vlan97
description ## Server  ##
ip address 192.168.97.254 255.255.255.0
!
!
```

Figure 38: vlan

4.2.3 Evaluation Outcome

The successful VLAN configuration verifies that network segmentation is correctly implemented, reducing broadcast traffic and enhancing security across departments.

4.3 Routing and NAT Evaluation

4.3.1 Purpose of the Test

Routing and NAT evaluation is performed to ensure that:

- Internal networks can communicate with each other
- Internal private networks can access the internet
- IP address translation is functioning correctly

4.3.2 Test Method

The following commands are executed on the router:

```
show Ip route
```

```
show Ip Nat translations
```

The routing table confirms that all VLAN subnets are reachable. The NAT translation table shows active translations between internal private IP addresses and the public IP address.

4.3.4 Evaluation Outcome

The results confirm that routing and NAT configurations are functioning as expected, allowing secure internet access for internal users while hiding private IP addresses.

4.4 Secure Shell (SSH) Access Evaluation

4.4.1 Purpose of the Test

SSH testing ensures that:

- Secure remote device management is enabled
- Unauthorized access via insecure protocols is prevented
- Network administrators can safely manage devices

4.4.2 Test Method

SSH login attempts are made from authorized management hosts using:

```
ssh -l admin
```

4.4.3 Test Results

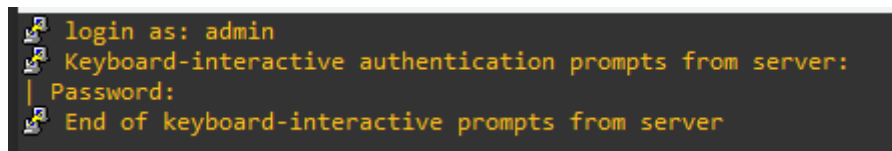


Figure 39: Successful SSH login session

The screenshot demonstrates a successful encrypted SSH connection to the network device using valid credentials.

4.4.4 Evaluation Outcome

The successful SSH login confirms that secure remote access is implemented, replacing insecure management protocols and improving overall network security.

4.5 Syslog Server Evaluation

4.5.1 Purpose of the Test

Syslog testing is conducted to verify that:

- Network devices are sending log messages to the centralized Syslog server
- Logs are recorded for monitoring and troubleshooting
- Security and system events can be audited

The logs display system messages, login records, and configuration events from network devices.

4.5.2 Evaluation Outcome

The presence of logs confirms that centralized logging is functioning correctly, enabling effective monitoring and faster troubleshooting.

4.6 DHCP Pool Utilization Evaluation

4.6.1 Purpose of the Test

DHCP evaluation ensures that:

- IP addresses are assigned dynamically
- Address pools are correctly configured
- IP address utilization is monitored

4.6.2 Test Method

The following command is executed on the DHCP server:

```
show Ip DHCP pool
```

4.6.3 Evaluation Outcome

The DHCP pool statistics confirm efficient IP address management and verify that sufficient addresses are available for current and future devices.

4.7 Redundancy and Failover Evaluation (HSRP)

4.7.1 Purpose of the Test

This test validates the redundancy mechanism to ensure continuous network availability during device failure.

4.7.2 Test Method

Failover is tested by disabling the primary router interface and observing traffic continuity through the standby router.

```
show standby brief
```

4.7.3 Evaluation Outcome

The successful failover demonstrates high availability and eliminates single points of failure in the network.

4.8 Summary of Evaluation Results

All evaluation tests confirm that the implemented network meets the design objectives. VLAN segmentation, routing, NAT, secure management, centralized logging, dynamic IP allocation, and redundancy mechanisms are functioning as expected. The results validate the effectiveness, security, and reliability of the Golden Hotel network implementation.

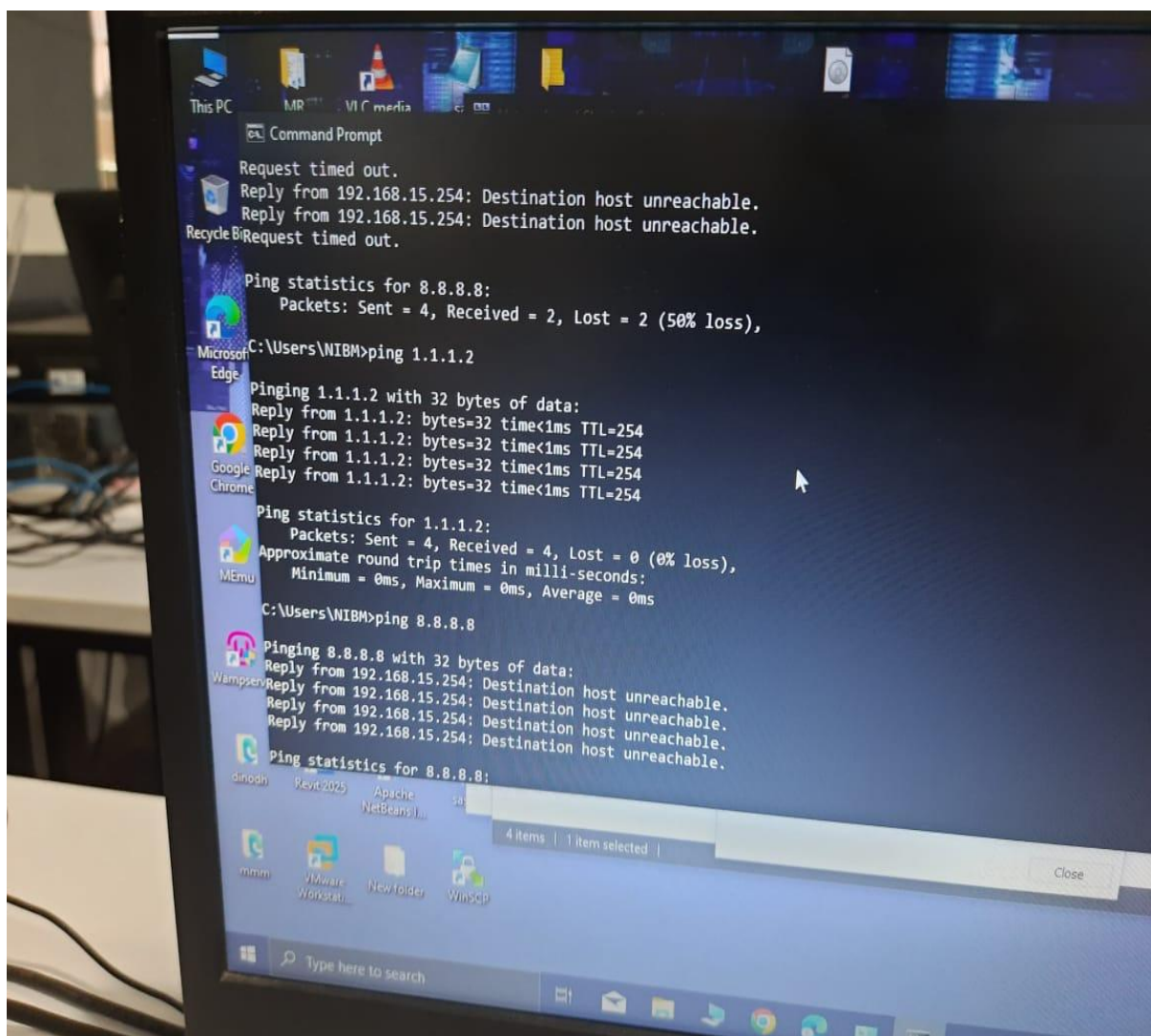


Figure 40: successful ping of the router port during implementation

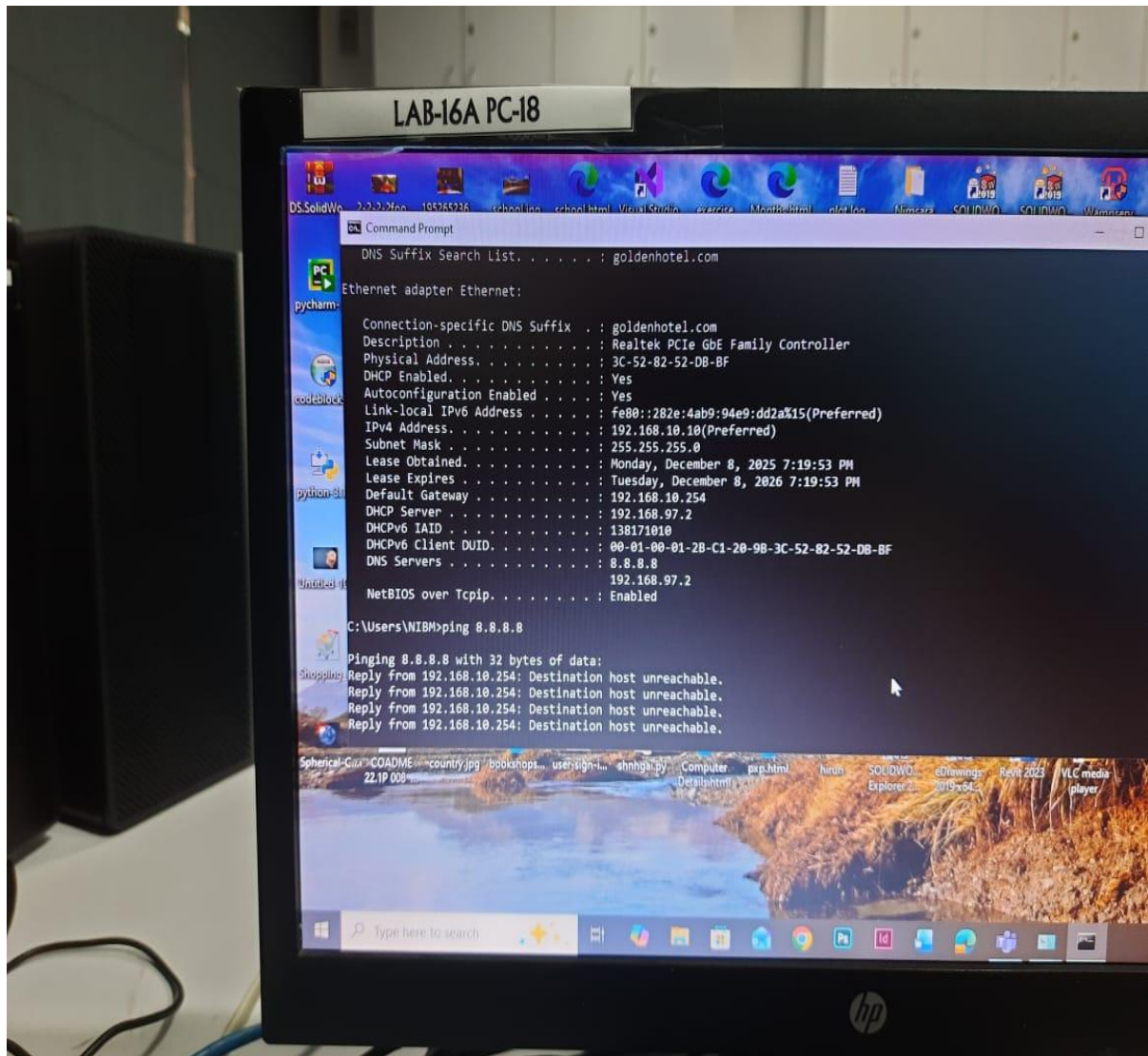


Figure 41: IP Addresses are obtained automatically with DHCP

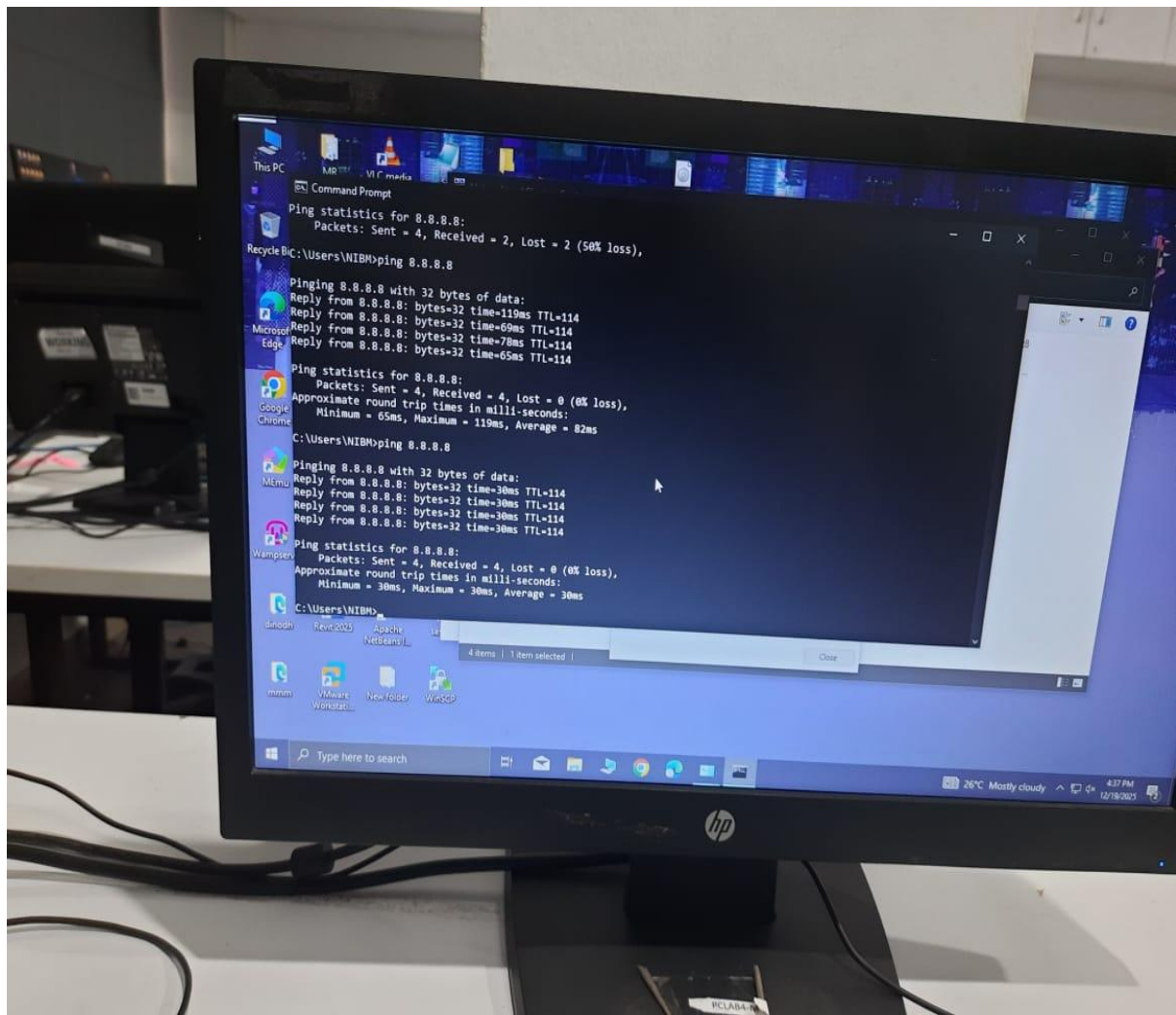


Figure 42: Successful ping to Google server

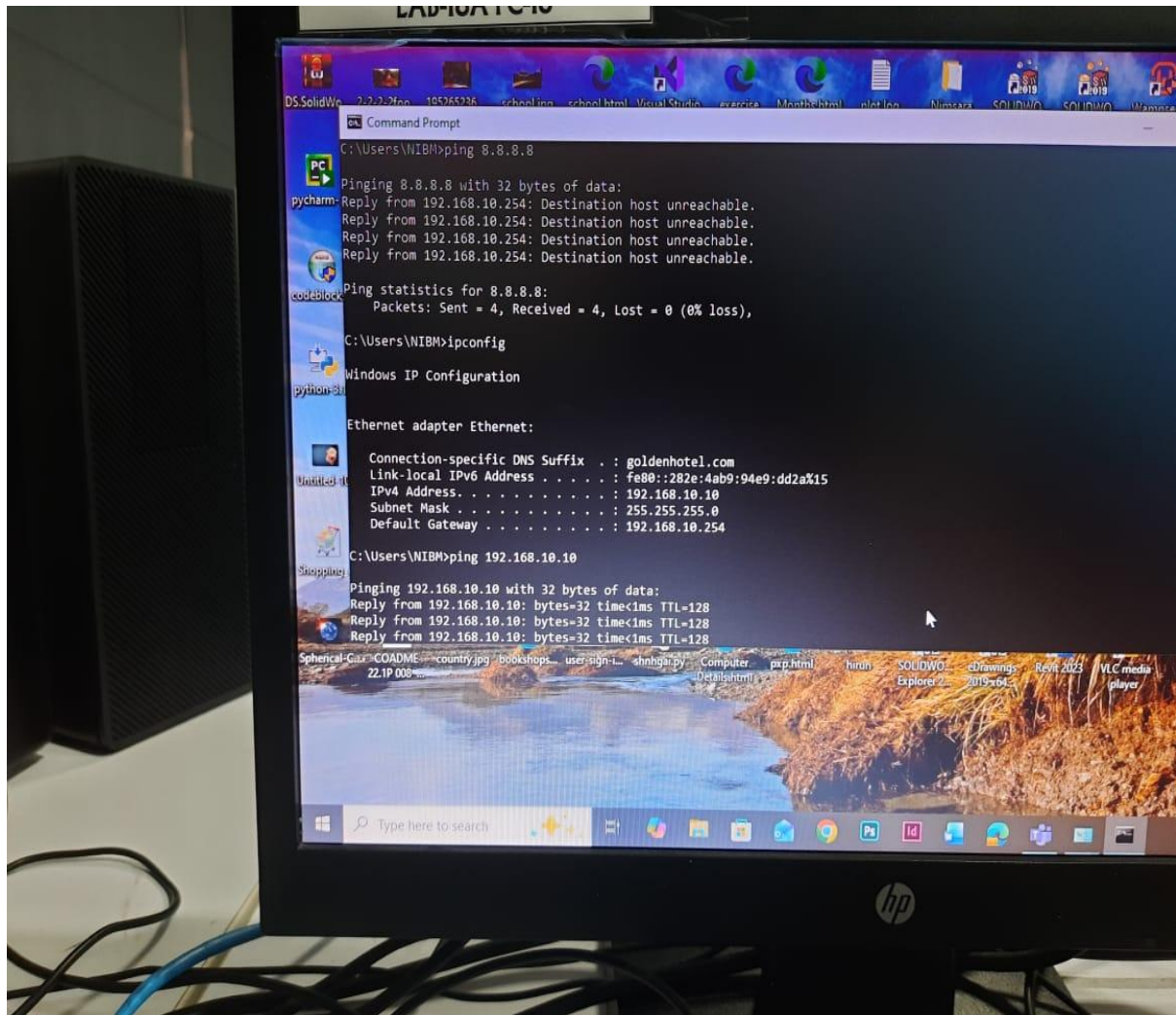


Figure 43: DHCP configured Successfully

CHAPTER 5 – CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project focused on the design, implementation, and evaluation of a secure and scalable computer network for Golden Hotel, a medium-scale hospitality organization with multiple departments and guest service areas. The primary objective of the project was to overcome the limitations of the existing unstructured network and to provide a reliable network infrastructure that supports daily hotel operations and future expansion.

Throughout the project, a structured approach was followed, beginning with an analysis of the organizational background and problem domain. Based on the identified challenges, a comprehensive network design was developed, incorporating best practices in enterprise networking. The design emphasized network segmentation, security, redundancy, and efficient resource management to meet the operational needs of a hotel environment.

The network was successfully implemented in a laboratory environment using industry-standard networking devices and technologies. Key features such as departmental network separation, dynamic IP address management, secure remote device access, centralized logging, and redundancy mechanisms were implemented and verified through systematic testing. The evaluation results confirmed that the implemented network operates as intended and satisfies the defined project objectives.

Overall, this project demonstrates how theoretical knowledge gained in computer networking can be effectively applied to solve real-world problems. The proposed network solution provides Golden Hotel with a robust, secure, and scalable network model that enhances operational efficiency, improves security, and ensures reliable service delivery for both staff and guests.

5.2 Achievement of Project Objectives

The objectives defined at the beginning of the project were successfully achieved as follows:

- A structured network design suitable for a hotel environment was developed.
- Network segmentation was implemented to reduce congestion and improve security.
- Dynamic IP address management simplified network administration.
- Redundancy mechanisms ensured high availability and minimized downtime.
- Secure remote management improved network administration security.
- Centralized monitoring enhanced troubleshooting and system visibility.
- The network was validated through comprehensive testing and evaluation.

The successful completion of these objectives confirms the effectiveness of the proposed network solution.

5.3 Future Work

Although the implemented network meets the current requirements of Golden Hotel, several enhancements can be introduced in the future to further improve performance, security, and manageability.

One potential enhancement is the integration of advanced firewall solutions to provide deeper packet inspection and intrusion prevention. This would significantly improve protection against modern cyber threats.

Another future improvement is the adoption of cloud-based services for data backup, monitoring, and centralized management. Cloud integration can enhance scalability and reduce dependency on on-site infrastructure.

Network automation and monitoring tools can also be introduced to simplify configuration management and enable proactive fault detection. Automation would reduce manual configuration errors and improve operational efficiency.

Additionally, expanding the wireless infrastructure with newer wireless standards and centralized wireless controllers would further enhance guest experience and support higher device density.

These future enhancements would allow Golden Hotel to maintain a modern and resilient network infrastructure capable of adapting to evolving technological requirements.

REFERENCES

All references used in this project follow the Harvard referencing style, which includes the author's name, year of publication, title, publisher, and source details.

- Sample References (Use Similar Format)
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ANNEXURE 1 – DEVICE SPECIFICATIONS

A1.1 Introduction

This annexure provides detailed specifications of the hardware devices used in the design and implementation of the Golden Hotel network. The selected devices are based on performance requirements, scalability, reliability, and suitability for a hotel environment.

A1.1.1 Layer 2 Switch - Cisco Catalyst C9300 (C9300-48P-E)



Figure 44: Cisco Catalyst C9300

The Cisco Catalyst C9300-48P-E is a secure, high-performance, enterprise-class stackable access switch. This switch is part of Cisco's Catalyst 9000 family and is built for technologies such as Wi-Fi 6, IoT, and cloud applications.

Port Configuration

- This switch has 48 Gigabit Ethernet PoE+ ports
- Network modules such as 4x1G, 4x10G, or 2x40G are available to support uplinks options.

Performance

- Providing transfer capacity of up to 128 Gbps.
- Providing a forwarding rate of 95.2 million packets per second (Mpps) for packet processing.

Stacking

- Simplified time management and support for stackwise-480 technology.
- Providing 480 Gbps stack bandwidth

Power supply

- Dual redundant, hot-swappable power supplies (to ensure uninterrupted operation and high availability)
- Supporting PoE+ power budget up to 740W

Software Features

- Cisco IOS XE, a modern modular operating system, helps deliver scalability, automation and advanced security.
- Integrating Cisco DNA Centres to enable intelligent and secure

A1.1.2 Firewall- Cisco Firepower 1120 (FPR1120-NGFW-K9)



Figure 45: Cisco Firepower 1120

The cisco Firepower 1120 is designed to provide security and connectivity for small and medium-sized enterprises and branch office and offers features such as URL filtering and software protection.

Hardware Specifications

- Featuring a redesigned multi-core CPU architecture for fast security processing and deep packet inspection.
- Providing 8x 1- Gigabit Ethernet RJ-45 interfaces for network connections and including 2 x SEP interfaces for uplinks or WAN connections.

Performance

- Delivering firewall throughput of up to 2 Gbps.
- Supporting threat throughput at 1.5 Gbps, ensuring robust security without reducing network speed.

Power supply

- Providing a 100-240V power supply for 24-hour operation.
- Supports redundant power via external UPS systems for high availability.

Software and Security Features

- Running on Cisco Firepower threat defence software.
- Includes advanced malware protection to managed user access and applications.

Management and Monitoring

- Supporting Cisco Security CDO for cloud-based policy management.
- Can be managed through Cisco Firepower Management Centre for centralized configuration and Monitoring.

A1.1.3 Access Point - Cisco Catalyst 9120AXI-E (C9120AXI-E)



Figure 46: Cisco Catalyst 9120AXI-E

The Cisco Catalyst 9120AXI-E, an enterprise-class Wi-Fi 6 wireless access point, is part of the Cisco Catalyst 9100 series, integrated with the Cisco DNA Centre for automation and analytics.

Hardware Specifications

- Model – Cisco Catalyst 9120AXI-E
- Form Factor – Indoor, Ceiling or wall mountable access point
- CPU – Quad-core ARM processor for efficient handling of wireless traffic
- Memory – 2048 MB DRAM and 1024 MB flash storage
- Antenna type – Internal omnidirectional antennas

Wireless Technology

- Support for Wi-Fi 6 standards, dual 4x4 MU-MIMO radios- 2.4 GHz and 5GHz
- Maximum data Rates
 - 2.4 GHz band- up to 575 Mbps
 - 5 GHz band – up to 4.8 Gbps
 - Aggregate throughput – up to 5.4 Gbps

Performance and Capacity

- Designed to support up to 1000 wireless clients per access point simultaneously, intelligent RF management automatically adjust channel and power for optimal coverage.
- Uses 802.11k, 802.11r and 802.11v protocols to provide seamless roaming between access points

Interfaces and Connectivity

- One 2.5 Gigabit Ethernet uplink port (supporting 100/1000/2500 Mbps speeds)
- One console port (RJ-45) for local management and configuration
- One USB 2.0 port for future use and diagnostics

A1.1.4 WLAN Controller- Cisco Catalyst 9800 (C9800-L-C-K9)



Figure 47: Cisco Catalyst 9800

The Cisco Catalyst 9800-L-C-K9 is a wireless controller designed for small and medium-sized enterprises and branch offices. It combines the reliability, security, and flexibility of next-generation Wi-Fi 6 and Wi-Fi 5 networks

Hardware Specifications

- From Factor – 1 RU rack-mountable appliance
- Processor – Multi- core CPU architecture optimized for wireless traffic processing
- Memory – 8 GB DRAM
- Flash Storage – 16 GB for system files and configurations
- Power supply – Dual redundant fixed power supplies for reliability

Wireless Capacity and Scalability

- Supports up to 250 access points, with the ability to support up to 5000 simultaneously on a wireless server.
- Designed for Wi-Fi 5 (802.11ac) and Wi-Fi 6 (802.11ax) access points.

Interfaces and Connectivity

- Eight 1-Gigabit Ethernet (RJ-45) ports, Two 10-Gigabit SFP+ uplink ports
- One RJ-45 console port and One USB 3.0 port

Performance and Reliability

- Providing seamless wireless operations with state transition between standby units and active.
- Minimize downtime by providing hardware and software redundancy.

Power and Environmental

- Input Voltage- 100–240V AC
- Typical Power Consumption- Approximately 100–120W (depends on usage)

A1.1.5 CCTV - Hikvision DS-9632NI-I8



Figure 48: Hikvision DS-9632NI-I8

The Hikvision DS-9632NI-I8 is a 32-channel professional network video recorder (NVR) designed for medium to large-scale surveillance systems. It is designed for 24-hour video surveillance.

Video and Recording Capabilities

- Supports up to 32 IP camera inputs, compatible with resolution up to 12 megapixels for ultra-HD video recording.
- Supports H.265+, H.265, H.264+, and H.264 video compression standards for efficient storage use.

Network and Connectivity

- Supports a wide range of bandwidth up to 320 Mbps
- Remote monitoring available through Hik-Connect, iVMS-4200, and web browsers.

Storage and Hard Drive Support

- Equipped with 8 SATA interfaces, supporting up to 10 TB per drive (maximum 80 TB total capacity).
- Supports RAID 0, 1, 5, 6, and 10 for data protection and redundancy.

Alarm and Event Management

- Includes 16 alarm inputs and 4 alarm outputs
- Includes line crossing, motion detection, video missing alert, and email notifications when events are deleted.

Interfaces and Ports

- Video Output- 2 x HDMI, 2 x VGA, 1 x CVBS
- Audio Input/Output- 16 x RCA inputs, 1 x RCA output (two-way audio supported)
- USB Ports- 2 x USB 2.0 (front) + 1 x USB 3.0
- Network Ports- 2 × RJ-45 (Gigabit)

Power and Environmental

- Power Supply- 100–240V AC, 50/60Hz

A1.1.6 Rack Server - Dell PowerEdge (R430 1U E5-2603)



Figure 49: - Dell PowerEdge

The Dell PowerEdge R430 is a 1U rack-mount server for small to medium-sized businesses and data centre's that require high-performance computing. It is best suited for web hosting, file sharing, and small- scale database applications.

Processor

- CPU - Intel Xeon E5-2603 v3 processor (6 cores, 6 threads)
- Base Clock Speed - 1.6 GHz
- Cache - 15 MB Intel Smart Cache
- Socket - LGA2011-3
- Thermal Design Power (TDP) - 85W

Memory

- Supports up to 384 GB DDR4 memory across 12 DIMM slots
- Compatible with DDR4 2133 MT/s memory modules

Storage and Drives

- Supports HDDs and SSDs for a mix of capacity and performance
- Maximum internal storage capacity - up to 48 TB (depending on configuration)

RAID and Storage Controller

- Supports PERC H330, H730, and H730P RAID controllers for redundancy and performance
- RAID levels supported - RAID 0, 1, 5, 6, 10, 50, 60

Network Connectivity

- Integrated dual-port 1GbE (Gigabit Ethernet) network interface controller
- Supports optional additional NICs (up to 10GbE) via PCIe expansion slots

Power Supply

- Supports redundant hot-plug power supplies – 350W, 450W, or 550W

A1.1.7 Layer 3 Switch – Cisco Catalyst 9500



Figure 50: Cisco Catalyst 9500

The Cisco Catalyst 9500 Series is a high-performance, modular, and scalable family of Layer-3 enterprise switches. The Cisco Catalyst 9500 Series is designed for campus core and network aggregation applications and offers high throughput, advanced routing, and rich Layer-3 services. The Cisco Catalyst 9500 Series also supports a comprehensive set of routing protocols and the programmable Cisco IOS XE software.

Performance & Scalability

- Switching Capacity – Up to 6.4 Tbps
- Forwarding Rate – Up to 2 Bpps
- MAC Addresses – Up to 82,000+ entries
- IPv4/IPv6 Routers – UP to hundreds of thousands
- VLANs – Supports 4,094 VLANs

Layer-3 & Protocol Capabilities

- Routing Protocols – OSPF, EIGRP, BGP, RIP, IS-IS
- IPv4/IPv6 hardware forwarding – for scalable dual-stick deployments
- Quality of Service – Advance classification, shaping, and policing
- Security – ACLs, TrustSec, MACsec
- Virtualization – VRF support for segmented routing

Hardware & Physical

- Chassis – 1 rack-unit fixed switch
- Weight – 9.5 – 11.6 kg with dual PSUs
- Environmental – Operating 0-40 °C
- MTU – Supports large frames for optimized L3 traffic

Power & Cooling

- Power Options – Dual, swappable (AC/DC) power supplies
- Typical PSU Ratings – 650W / 930W / 1600W or DC options
- Airflow – Front-to-back or back-to-front depending on development
- Fans – Field-replaceable modular fans with redundancy support

Management & Software

- Cisco IOS XE – Industry-standard modular OS with automation APIs
- Management – Supports CLI, SNMP, Cisco DNA Center
- Security & Compliance – Secure boot, role-based access, SSH, RADIUS, TACACS+

A1.1.8 Layer 1 switch – Cisco Catalyst 9200



Figure 51: Cisco Catalyst 9200

The Cisco Catalyst 9200 Series is another important switch. Designed for secure, resilient campus access and aggregation applications, this switch is a fully configurable, Layer-3 managed switch that runs on the Cisco IOS XE operating system.

Model & Ports

- Model – Cisco Catalyst 9200
- Port Configuration
 - 24 x 10/100/1000 Mbps RJ-45 Ethernet ports for access connectivity
 - Supports modular 1G/10G/25G/40G uplinks for aggregation or core networks.

Performance & Scalability

- Switching Capacity - 128 Gbps
- Forwarding Rate – 95.23 Mpps
- Stacking Bandwidth – Stackwise- 160 for multi-switch high-availability stacks
- MAC Addresses – Up to 32,000 address entries

Software & Management

- Cisco IOS XE - Modular OS with support for automation, programmability, and telemetry.
- Licensing - Network Advantage, enabling advanced routing and feature sets.
- Management - Supports CLI, SNMP, NetFlow, Cisco DNA Center, and telemetry APIs.

Power & Environmental

- Power Supply - Default AC power module
- Fan Modules - Field-replaceable, redundant-capable
- Operating Temperature - 0 °C–45 °C
- Relative Humidity- 5–90 % non-condensing.

Typical Use Cases

- Enterprise Access - 1G edge connectivity for PCs, IP phones, and wireless Aps
- Stacked Access - Scalable and resilient campus deployment using stacked switches
- Routed Access - Supports IPv4/IPv6 for modern segmented campus networks.

A1.1.9 Types of Network Cables

1. Twisted Pair Cables

Twisted pair cables are the most common type of cabling used in local area networks (LANs) and other telecommunications setups. They consist of pairs of insulated copper wires twisted together to reduce electromagnetic interference.

Unshielded Twisted Pair (UTP)

This is the most used twisted pair cable in office and home networks. It relies on the twisting of the cables to provide protection from electromagnetic interference.

Shielded Twisted Pair (STP)

This cable includes shielding in the form of a foil around each pair of wires or around the group of wires to provide additional protection from interference.

2. Coaxial Cable

Coaxial cable, or coax, is made of a single solid copper core. A plastic layer provides insulation between the center conductor and a metal shield. This cable is traditionally used for television and other broadband networks due to its ability to carry high-frequency signals over long distances without significant loss.

3. Fiber Optic Cable

Fiber optic cables use glass fibers to transmit data as pulses of light. They offer high bandwidth, can support data transmission over long distances, and are immune to electrical interference. Fiber optic cables are ideal for backbone connections between networked buildings or within data centers.

Single-mode Fiber

Uses a single ray of light (mode) to carry transmission over long distances.

Multi-mode Fiber

Uses multiple rays of light simultaneously, which gives high bandwidth at high speeds over medium distances.

For more detailed information, refer to [Advantages of Using Fiber Optic Cabling for a Network](#).

4. Ethernet Crossover Cable

A crossover cable is a type of twisted pair cable that reverses the transmit and receive signal pins. It is used to connect two similar devices directly, such as a computer to a computer or a switch to a switch without needing a switch or hub in between.

5. Console cables

also known as Cisco cables, rollover cables and management cables — are designed for a specific purpose. They connect Cisco networking devices to terminals or PCs for configuration. Typically, the Cisco end will connect via RJ45, and the terminal end will conclude in a serial connection.



Figure 52: Console cable

A1.1.10 Desktop PC – Lenovo Yoga AIO 7

The Lenovo Yoga AIO 7 is a 27-inch all-in-one desktop computer that features a large 4K screen along with impressive capabilities for artistic, entertainment, and business purposes. A great feature about it is that you can position it in flexible positions that would allow you to adjust the screen in whatever position you prefer.

Display

- 27" 4K UHD (3840 × 2160) IPS panel
- Touchscreen available on some models
- can rotate the display easily

Processor

- AMD Ryzen™ 5 6600H or Ryzen™ 7 6800H mobile CPUs

Graphics

- Up to **AMD Radeon™ RX 6600M** discrete GPU (8 GB)

Memory & Storage

- **RAM:** 16 GB or 32 GB LPDDR5 (soldered)
- **Storage:** SSD options — 256 GB / 512 GB / 1 TB (PCIe NVMe)
- Support for up to two SSDs

Ports

- 2x USB-C, 4x USB-A, HDMI-out, Ethernet, SD card reader



Figure 53: Desktop PC

A1.1.11 Laptops – Lenovo ThinkPad X1 Carbon Gen11

The Lenovo ThinkPad X1 Carbon Gen 11 is a high-end offering in the 14-inch category of business ultra books, catering to professionals who demand a light, tough, and high-performance laptop computer. The device comes from Lenovo's ThinkPad X1 lineup of laptops, which are well-known for its sturdy quality and keyboard

Display

- 14.0" screen
- Typical resolution: 1920 × 1200 (WUXGA) IPS

Processor

- 13th Gen Intel Core™ options

Graphics

- Intel Iris™ Xe integrated graphics

Memory

- Up to 64 GB LPDDR5/x

Storage

- Single M.2 PCIe 4.0 SSD up to 2 TB



Figure 54: Laptops

A1.1.12 IP Phone – Cisco SPA 303

The Cisco SPA 303 is a 3-line IP phone that is SIP-enabled for effective VoIP communication. This device is equipped with basic business communication features such as multi-line functionality, caller ID, and conferencing capabilities without spending an arm and a leg for such functionality

Design & Hardware

- Compact, 2-line LCD
- 3 programmable line keys
- Built-in Ethernet switch

Call Management & Features

- Call Hold, Transfer, Conferencing, Redial
- Caller ID, Missed Call Indicator
- Speed Dial

Connectivity & Compatibility

- SIP compliant
- Dual 10/100 Mbps Ethernet ports

Power Options

- Power over Ethernet (PoE) 802.3af
- Optional AC adapter



Figure 55: Ip Phones

A2.2 Layer 3 Switch Specifications

Feature	Specification
Device Type	Layer 3 Managed Switch
Ports	24/48 Gigabit Ethernet
VLAN Support	Yes
Inter-VLAN Routing	Supported
Redundancy	HSRP capable
Security	ACL support
Management	Secure remote access

Table 2: Layer 3 Switch Specifications

A2.3 Layer 2 Switch Specifications

Feature	Specification
Device Type	Layer 2 Managed Switch
Ports	24/48 Gigabit Ethernet
VLAN Support	Yes
Port Security	Supported
Trunking	Supported
Management	Secure remote access

Table 3: Layer 2 Switch Specifications

A2.4 Wireless Access Point Specifications

Feature	Specification
Wireless Standard	Modern wireless standard
Frequency Bands	Dual-band
SSID Support	Multiple SSIDs
Security	Encrypted wireless access
Deployment	Ceiling or wall mounted

Table 4: Wireless Access Point Specifications

A2.5 Server Hardware Specifications

Feature	Specification
Processor	Multi-core CPU
Memory	16–32 GB RAM
Storage	SSD / HDD
Virtualization Support	Enabled
Network Interfaces	Gigabit Ethernet
Purpose	DHCP, Syslog, management services

Table 5: Server Hardware Specifications

A2.6 Rack and Power Equipment

Feature	Specification
Network Rack	19-inch standard rack
Power Backup	UPS
Cooling	Adequate ventilation
Cable Management	Structured cable organizers

Table 6: Rack and Power Equipment

A2.7 Summary

The selected hardware devices provide sufficient performance, reliability, and scalability to support the network requirements of Golden Hotel. These specifications ensure that the network can handle current operational demands and future expansions.

BUDGET ANALYSIS

- Hardware cost
- Software cost
- Total project cost

Device name	Product ID	Unit Price	Quantity	Total Price
Layer-3 Switch	C9500-48Y4C	LKR 6,806,925	2	LKR 13,613,850
Layer 2 Switch	C9300 – 48P - E	LKR. 3,495,080	2	LKR.6,990,161
Layer 1 Switch	C9200 – 24P - E	LKR. 556,930	7	LKR. 3,898,510
Firewall	FPR1120 – NGFW – K9	LKR. 1,716,464	1	LKR.1,716,464
Access point	C9120AXI - E	LKR. 667,599	66	LKR. 44,061,534
WLAN Controller	C9800 - L – C – K9	LKR. 3,648,143	1	LKR.3,648,143
ISE Licensing	250 End point 1 year Package	-	250	LKR. 525,989
NVR	DS – 9632NI – I8	LKR. 292,828	1	LKR.292,828
Cables	-	-	-	LKR.200,000
Servers for DHCP/DNS/PM S/Syslog	Dell PowerEdge R430 1U E5-2603	LKR. 531,544	1	LKR.531,544
Total Budget				LKR. 75,482,023

Table 7: ESTIMATED BUDGET

A2.8 Summary of Project Outcomes

This project successfully achieved its primary objective of designing and implementing a secure, scalable, and reliable computer network for Golden Hotel. A structured network architecture was developed to support hotel operations across multiple departments and guest areas.

The implemented network provides effective separation between internal departments and guest access, improving both performance and security. Centralized management of network services simplifies administration, while redundancy mechanisms enhance availability and minimize downtime. Secure access controls and monitoring mechanisms contribute to improved protection of sensitive hotel systems.

Overall, the project outcomes demonstrate that the proposed network solution effectively addresses the identified problems and meets the operational requirements of a modern hotel environment. The laboratory implementation and evaluation results confirm the feasibility and reliability of the designed network.

A2.9 Lessons Learned

During the execution of this project, several valuable lessons were learned, both technical and practical.

From a technical perspective, the project enhanced understanding of network design principles, including segmentation, redundancy, and security implementation. Practical configuration of networking devices improved skills in troubleshooting, verification, and performance evaluation.

From a project management perspective, the importance of proper planning, documentation, and time management became evident. Designing before implementation helped reduce configuration errors and ensured smoother execution. Testing and evaluation highlighted the need for systematic validation of each network component.

Additionally, the project emphasized the importance of aligning theoretical knowledge with real-world requirements. This experience has strengthened problem-solving abilities and provided practical exposure to enterprise-level networking environments.